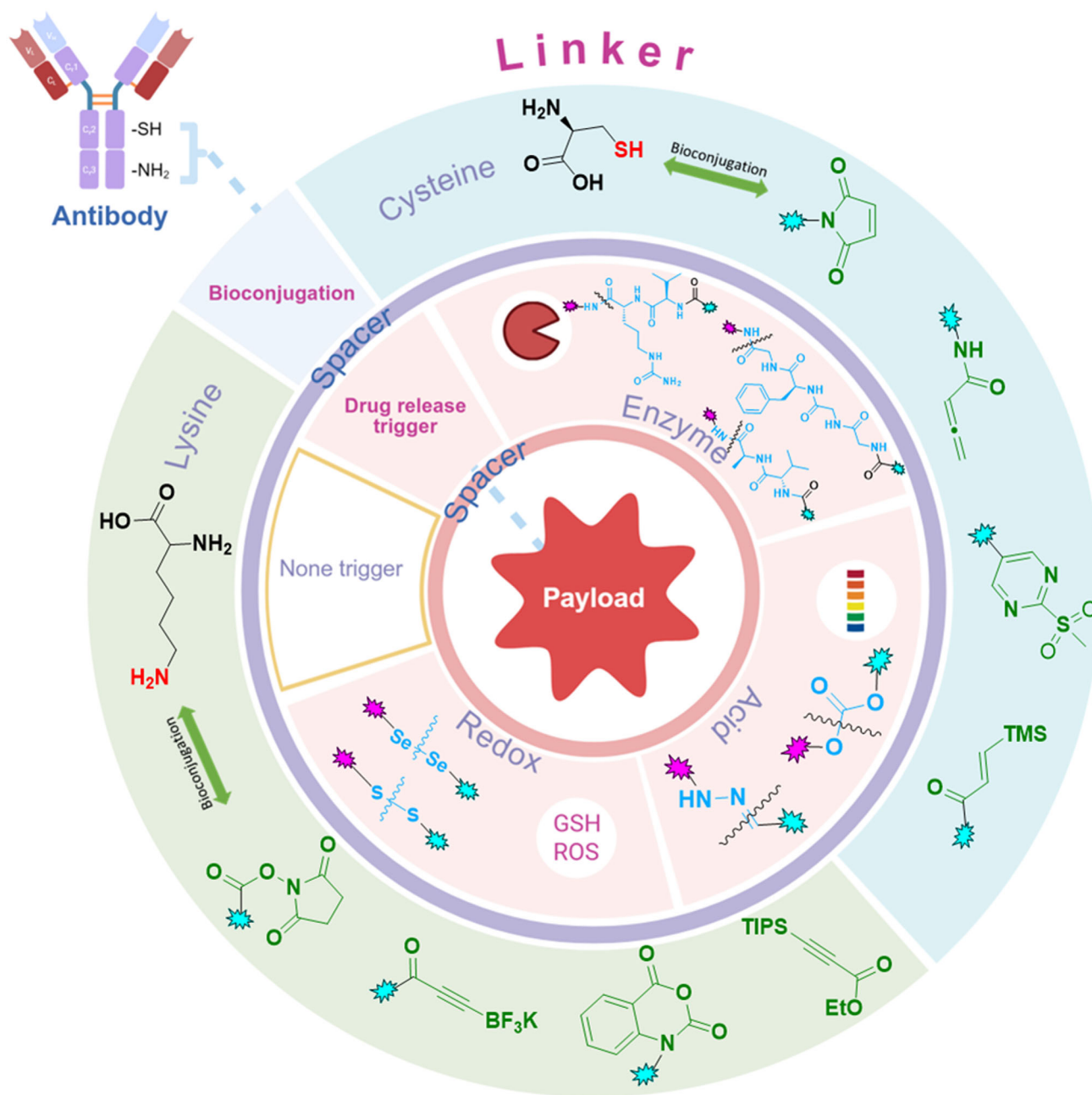


Linker Design for the Antibody Drug Conjugates: A Comprehensive Review

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Inspired by the “magic bullet” concept proposed over a century ago, antibody–drug conjugates (ADCs) are developed to enhance cancer therapy by linking monoclonal antibodies to a cytotoxic payload, aiming to overcome the limitations of conventional chemotherapy. To date, 17 ADCs have received regulatory approval for treating both hematologic and solid tumors. Despite their clinical success, developing ADCs with optimal therapeutic potential remains challenging. While selecting the appropriate antibody and cytotoxin is crucial, the linker plays a pivotal role in determining plasma stability and efficient

payload release at the tumor site. Over the past decade, advances in linker technology have significantly improved the pharmacokinetics, efficacy, and toxicity profiles of ADCs. This review provides an overview of clinically validated linkers and recent innovations in linker design, focusing on drug release triggers, bioconjugation strategies, the impact of spacers on hydrophilicity, traceless drug release, and linker architecture, as well as a discussion of the bystander effect, offering insights for the rational design of next-generation ADCs.

1. Introduction

Antibody–drug conjugates (ADCs), designed to combine the selectivity of monoclonal antibodies with the potent cytotoxic effects of chemotherapy drugs, have emerged as a powerful tool in cancer treatment. ADC technology aims to maximize therapeutic efficacy while minimizing off-target toxicity, a significant limitation of conventional chemotherapy. The concept of the “magic bullet”—a targeted therapeutic agent capable of selectively delivering a toxin to a pathogen—was first proposed by Paul Ehrlich, the Nobel Prize winner in Physiology or Medicine in 1908.^[1] Nearly nine decades later, in 2000, gemtuzumab ozogamicin (Mylotarg) became the first ADC approved for the treatment of relapsed acute myeloid leukemia.^[2] Since then, ADCs have become a rising star in cancer treatment within the pharmaceutical industry, with a rapidly increasing number of clinical studies from 2000 to 2024 (Scheme 1). To date, 17 ADCs have received regulatory approval, covering indications for both hematologic malignancies and solid tumors: gemtuzumab ozogamicin (Mylotarg, 2000/2011),^[2] brentuximab vedotin (Adcetris, 2011),^[3] adotrastuzumab emtansine (Kadcyla, 2013),^[4] inotuzumab ozogamicin (Besponsa, 2017),^[5] polatuzumab vedotin (Polivy, 2019),^[6] moxetumomab pasudotox (Lumoxiti, 2018),^[7] enfortumab vedotin (Padcev, 2019),^[8] trastuzumab deruxtecan (Enhertu, 2019),^[9] sacituzumab govitecan

(Trodelvy, 2020),^[10] belantamab mafodotin (Blenrep, 2020),^[11] cetuximab sarotolacan (Akalux, 2020),^[12,13] loncastuximab tesirine (Zynlonta, 2021),^[14] disitamab vedotin (Aidixi, 2021),^[15] tisotumab vedotin (Tivdak, 2021),^[16] mirvetuximab soravtansine (Elahere, 2022),^[17] sacituzumab tirumotecan (Jiatailai, 2024),^[18] and datopotamab deruxtecan (Datroway, 2025).^[19] The success of these ADCs highlights their potential to revolutionize cancer treatment by offering a more targeted and effective alternative to conventional chemotherapy.^[20,21]

ADCs comprise three core components: 1) a monoclonal antibody specifically targeting tumor-associated antigens, 2) a cytotoxic payload (warhead), and 3) a chemical linker connecting them.^[22] Unlike non-targeted therapies, ADCs selectively bind to cancer cell surface antigens, concentrating cytotoxic payloads at tumor sites while minimizing systemic toxicity. Upon antigen binding, the ADC–antigen complex undergoes endocytosis, followed by intracellular linker degradation, leading to payload release. There are also other mechanisms for certain ADCs to release the payload without antigen engagement and internalization, as reviewed by Tolaney et al. in 2023.^[23] The released payload induces cell death via diverse mechanisms: microtubule inhibitors (e.g., monomethyl auristatin E [MMAE]/monomethyl auristatin F [MMAF]/Mertansine [drug maytansinoid (D1)/Ravtansine (DM4)] disrupt cytoskeletal dynamics;^[17,24–26] DNA-damaging agents (e.g., calicheamicin ozogamicin, pyrrolbenzodiazepine [PBD] dimer SG3199) cause DNA breaks;^[2,5,14] topoisomerase I inhibitors (e.g., SN-38/exatecan derivative Dxd/belotecan derivatives) impair DNA replication;^[9,10,18] immune modulators (e.g., T785) stimulate immune responses;^[27] photosensitizers (e.g., IRDYE700) enable photodynamic therapy;^[12,13] and protein synthesis inhibitors (e.g., PE38) block protein synthesis.^[7]

As the molecular bridge between the antibody and payload, the linker plays a pivotal role in ADC stability, targeting precision, and therapeutic efficacy. An optimally designed linker ensures controlled drug release at the tumor site while maintaining systemic stability to prevent premature payload release. Structurally, ADC linkers consist of two essential components (Scheme 1): the drug release trigger, which includes a suitable spacer for attachment to the payload; and the bioconjugation group responsible for linking to the side chain of specific amino acid (AA) residues of the antibody. In addition, there are spacers serving as junctions between the two parts or between the trigger and payload.^[28] Spacers can be tuned to enhance linker stability, increase hydrophilicity,^[14] and achieve the traceless release of the payload following the

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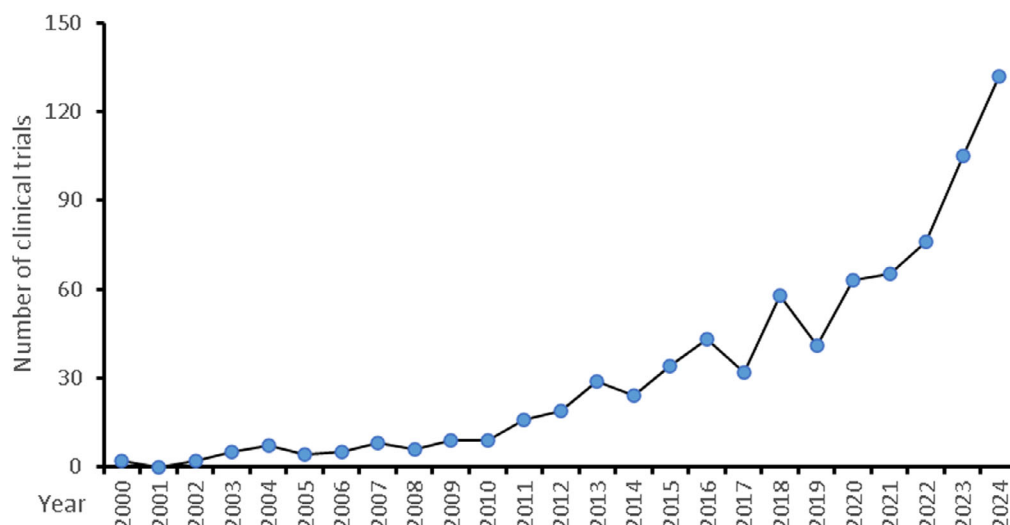
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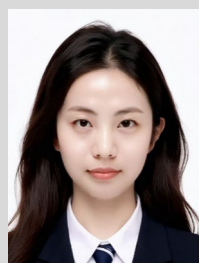
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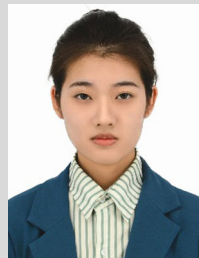
Scheme 1. Number of registered clinical trials on antibody drug conjugates recorded in clinicaltrials.gov from 2000 to 2024. Search parameter (disease “cancer”, other term “antibody drug conjugates”, date type “study start”) was set and optimized by following work by Guo et al.^[166]

processing of the trigger in response to desired conditions.^[29,30] Depending on whether a drug release trigger is present in the linker, serving as a conditional cleavage site, the linkers are categorized into cleavable and non-cleavable linkers.^[31]

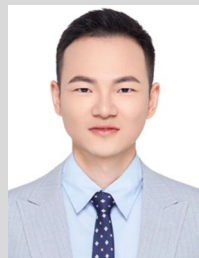
This review provides a comprehensive analysis of linker design by summarizing the structural characteristics of the 17 ADCs approved by regulatory agencies and highlighting recent advancements in drug release triggers, bioconjugation, and



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spacer technology with potential applications in linker development. These insights aim to inform the rational design of next-generation ADCs and contribute to further advancements in the field.

2. Drug Release Trigger

The drug release trigger of the linker determines whether the drug release pattern is triggerable or non-triggerable. In cleavable linkers, a chemical drug release trigger facilitates payload release upon exposure to specific biological stimuli, such as lysosomal enzymes, acidic pH, or redox conditions.^[32] In contrast, non-cleavable linkers lack such a trigger and function solely as stable conjugation adaptors that tether the payload to the antibody. Among the 17 ADCs approved for clinical use, there are 13 with cleavable linkers bearing a drug release trigger: 8 with enzyme-labile triggers, 2 with acid-labile triggers, 1 with a redox-labile trigger, and 2 with both acid- and redox-labile triggers (Table 1).

2.1. Enzyme-Liable Drug Release Trigger

Enzyme-labile drug release triggers are well established in ADC design. These triggers incorporate peptide motifs that are selectively recognized and cleaved by enzymes overexpressed in pathological tissues, such as tumors, promoting targeted and controlled payload release.

As shown in Scheme 3A–C, eight approved ADCs utilize enzyme-labile triggers, including brentuximab vedotin,^[24] polatuzumab vedotin,^[33] enfortumab vedotin,^[34] disitamab vedotin,^[15] tisotumab vedotin,^[16] loncastuximab tesirine,^[14] trastuzumab deruxtecan,^[9] and datopotamab deruxtecan.^[35] Brentuximab vedotin, polatuzumab vedotin, enfortumab vedotin, disitamab vedotin, and tisotumab vedotin share the same linker–payload structure. The linker, named **Mc–Val–Cit–PABC** (*p*-aminobenzyloxycarbonyl [PABC]), includes a thiol-reactive maleimidocaproyl (MC) group for cysteine bioconjugation, a protease-sensitive valine–citrulline (Val–Cit or VCit) dipeptide as a drug release trigger, and a self-immolative PABC spacer (Scheme 2). The PABC spacer between the trigger and payload enables the traceless release of MMAE (a tubulin polymerization inhibitor) upon cleavage and facilitates the enzyme recognition and cleavage of the VCit dipeptide. Each ADC targets distinct tumor-associated antigens: brentuximab vedotin targets CD30 with a half-life 4.4–5.9 days,^[36] polatuzumab vedotin targets CD79b with a half-life 4.43–7.98 days,^[37] enfortumab vedotin targets Nectin 4 with a half-life 3.4 days,^[8] disitamab vedotin targets human epidermal growth factor receptor 2 (HER2) with a half-life 1–1.5 days,^[38] and tisotumab vedotin targets tissue factor with a half-life 4.04 days (2.26–7.25 days).^[39] Loncastuximab tesirine consists of a humanized anti-CD19 antibody linked to SG3199, a PBD payload, via a linker named **Mal–PEG8–Val–Ala–PABC** that is cleavable in response to cathepsin (Scheme 3B).^[14,40] This linker contains a thiol-reactive maleimide (Mal) group for cysteine bioconjugation, a hydrophilic polyethylene glycol (PEG) spacer, a protease-sensitive valine–alanine (Val–Ala) dipeptide as the drug release trigger. It also

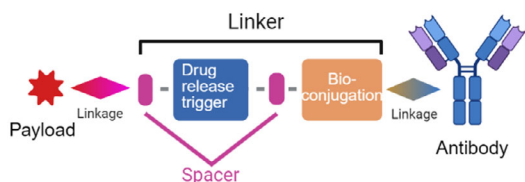
includes a self-immolative PABC spacer for the traceless release of the payload SG3199, a synthetic PBD dimer that acts as a cytotoxic DNA minor-groove interstrand cross-linking reagent. This drug has a half-life 8.85 days after first dose and 15.2 days after the second dose.^[41] Trastuzumab deruxtecan employs an anti-HER2 antibody conjugated to Dxd (a topoisomerase I inhibitor) via the linker **Mc–Gly–Gly–Phe–Gly–AM** (Scheme 3C) with a half-life 5.8 days.^[9] This linker has a thiol-reactive MC group for cysteine bioconjugation, a protease-sensitive glycine–glycine–phenylalanine–glycine (GGFG) tetrapeptide as the drug release trigger, and a self-immolative aminomethyl (AM) spacer for the release of the payload Dxd, a potent DNA topoisomerase I inhibitor through amino elimination. The tetrapeptide GGFG trigger is recognized by lysosomal enzymes (e.g., cathepsin B and L).^[42] Datopotamab Dxd has the same linker–payload system as trastuzumab deruxtecan, with the antibody directed to trophoblast cell surface antigen 2 (TROP2) and half-life 4.8 days.^[43,44]

The cathepsin enzymes responsible for cleavage in these ADCs belong to the lysosomal cysteine protease family, including cathepsins B, H, and L. Cathepsin B is frequently overexpressed in tumors and is associated with cancer invasiveness and metastasis.^[45] VCit-based linkers are designed with peptide sequences recognized by cathepsin B, ensuring efficient ADC cleavage in lysosomes after cellular internalization (Scheme 3D). However, despite its widespread use, the VCit-based linker exhibits instability in mouse plasma, attributed to Ces1c, an extracellular carboxylesterase.^[46] This instability complicates preclinical drug development, leading to misleading stability and safety profiles in mouse models. To address this, researchers have explored novel cathepsin B-sensitive linkers with enhanced stability. In 2018, Tsuchikama et al. reported an EVCit (glutamic acid–valine–citrulline) tripeptide linker that extends the half-life (*t*_{1/2}) from 2.3 h (VCit) to 19.1 h (Scheme 3E).^[47] The same group reported in 2022 another EGCit (glutamic acid–glycine–citrulline) trigger with improved stability and reduced adverse effects, such as neutropenia and hepatotoxicity (Scheme 3F).^[48] Alhamadsheh et al. in 2022 demonstrated that using the EVCit tripeptide in a prostate-specific membrane antigen-targeting small-molecule drug conjugate resulted in superior stability (71.3% vs 20.9% remaining in mouse serum after 24 h) and reduced toxicity, as evidenced by less weight loss compared to the VCit conjugate.^[49] In addition to EVCit and EGCit, other trigger designs have been reported with higher specificity to cathepsin B. Li et al. in 2016 designed a triglycyl peptide (CX)-based linker, cleaved in lysosomes without requiring cathepsin B.^[50] This linker achieved pharmacokinetics (*t*_{1/2}, CL, AUC_{0–∞}) comparable to N-succinimidyl-4-(N-maleimidoethyl) cyclohexane-1-carboxylate (SMCC)-based non-cleavable linkers and had 5–100-fold lower IC₅₀ values than non-cleavable linkers, with a remarkable therapeutic index. As reviewed with EVCit trigger by Gennari et al. in 2019,^[51] Safina et al. in 2018 developed a cyclobutane-1,1-dicarboxamide (cBu)-modified linker (cBuCit), which achieved 75% lower levels of payload (MMAE) in plasma compared to the VCit linker (Scheme 3G).^[52] In 2024, Tai et al. introduced a mono-AA linker for controlled release of MMAF.^[26] Unlike traditional designs, the N-terminal of the AA is directly conjugated to the C terminal of phenylalanine in MMAF. Screening various AAs (Ala, Ahx, Asp, Lys, and Asn) revealed that

Table 1. Information on the 17 approved ADCs. Drug release triggers: 8 with enzyme-labile triggers, 2 with acid-labile triggers, 1 with a redox-labile trigger, 2 with both acid- and redox-labile triggers, 3 without triggers, and 1 without chemical linker. Bioconjugation: 11 are manufactured using cysteine-dependent bioconjugation and 5 are manufactured using lysine-dependent bioconjugation.

ID	Brand name	Common name	Antibody target	mAb	Payload	Payload action	Linker	Drug release trigger	Bioconjugation	Drug-to-Antibody Ratio (DAR)	Terminal half-life (day)	Year	Reference
1	Adcetris	Brentuximab vedotin	CD30 (TNFSF8)	IgG1	MMAE	Microtubule inhibitor	Mc-Val-Cit-PABC	Enzyme (cathepsin)	Cysteine	4	4.4–5.9	2011	3, 37
2	Polivy	Polatuzumab vedotin	CD79b	IgG1	MMAE	Microtubule inhibitor	Mc-Val-Cit-PABC	Enzyme (cathepsin)	Cysteine	3.5	4.43–7.98	2019	6, 38
3	Padcev	Enfortumab vedotin	Nectin4	IgG1	MMAE	Microtubule inhibitor	Mc-Val-Cit-PABC	Enzyme (cathepsin)	Cysteine	3.8	3.4	2019	8
4	Aidixi	Disitamab vedotin	HER2	IgG1	MMAE	Microtubule inhibitor	Mc-Val-Cit-PABC	Enzyme (cathepsin)	Cysteine	4	1–1.5	2021	15, 39
5	Tivdak	Tisotumab vedotin	Tissue factor	IgG1	MMAE	Microtubule inhibitor	Mc-Val-Cit-PABC	Enzyme (cathepsin)	Cysteine	4	4.04 (2.26–7.25)	2021	16, 40
6	Enhertu	Trastuzumab deruxtecan	HER2	IgG1	Dxd	TOP1 inhibitor	Mc-Gly-Gly-Phe-Gly-AM	Enzyme (cathepsin)	Cysteine	8	5.8	2019	9
7	Datroway	Datopotamab deruxtecan	TROP2	IgG1	Dxd	TOP1 inhibitor	Mc-Gly-Gly-Phe-Gly-AM	Enzyme (cathepsin)	Cysteine	4	4.8	2025	19, 44
8	Zynlonta	Loncastuximab tesirine	CD19	IgG1	SG3199	DNA cleavage	Mal-PEG8-Val-Ala-PABC	Enzyme (cathepsin)	Cysteine	2.3	8.85	2021	14, 42
9	Mylotarg	Gemtuzumab ozogamicin	CD33	IgG4	Ozogamicin	DNA cleavage	AcBut	Acid/Redox (GSH)	Lysine	2–3	3 ± 1.75	2000	2, 68
10	Besponsa	Inotuzumab ozogamicin	CD22	IgG4	Ozogamicin	DNA cleavage	AcBut	Acid/Redox (GSH)	Lysine	6	12.3	2017	5, 69
11	Trodelyv	Sacituzumab govitecan	TROP2	IgG1	SN-38	TOP1 inhibitor	CL2A	Acid	Cysteine	7.6	0.48–0.78	2020	10, 70
12	Jlaltalai	Sacituzumab tirumotecan	TROP2	IgG1	Belotecan-derivative	TOP1 inhibitor	Pyrimidine-CL2A-carbonate	Acid	Cysteine	7.4	2.34	2024	18, 72
13	Elahere	Mirvetuximab soravtansine	FRα	IgG1	DM4	Microtubule inhibitor	Sulfo-SPDB	Redox (GSH)	Lysine	3.5	4.8–5.8	2022	17, 90
14	Kadcyla	Trastuzumab emtansine	HER2	IgG1	DM1	Microtubule inhibitor	SMCC	None	Lysine	3.5	3.5	2013	4, 102
15	Blenrep	Belantamab mafodotin	BCMA	IgG1	MMAF	Microtubule inhibitor	Maleimido-caproyl	None	Cysteine	4	11.5	2020	100 101
16	Akalux	Cetuximab saratolacan	EGFR	IgG1	IRDY700	Membrane damage	Linear alkyl/alkoxy linker	None	Lysine	1.3–3.8	n/a	2020	12, 13
17	Lumoxiti [®]	Moxetumomab pasudotox	CD22	IgG1	PE38	Protein synthesis inhibition	N/A	N/A	N/A	2	0.058 (1.4 h)	2018	7, 107

^a Moxetumomab pasudotox is an ADC manufactured by expression in bacteria without a chemical linker.^[7]



Scheme 2. The typical structure of an ADC linker includes a drug release trigger, a bioconjugation group, and spacers. The drug release trigger determines how the payload is released in response to different conditions, such as enzymes or acidic environments. The bioconjugation group is responsible for attachment to the antibody. Spacers between the trigger and payload influence whether the drug release is traceless, while spacers between the trigger and the bioconjugation group serve as junction and can impact linker parameters like hydrophilicity. Linkers with drug release trigger are classified as a cleavable linkers. Otherwise, linkers that include only a bioconjugation group and spacers for attachment to the payload are categorized as non-cleavable linkers. Created in BioRender.

the Asn-linker exhibited the highest potency and fastest cleavage by cathepsin B. The cleavage occurred at the C terminal of Asn, releasing an Asn-prodrug, which was subsequently processed by exopeptidase to yield active MMAF. Although this trigger appears to be drug specific, it is a feasible strategy for peptide-based payloads like MMAF. Future research is needed to optimize linker stability, pharmacokinetics, and payload release efficiency to enhance ADC therapeutic efficacy while minimizing off-target toxicity.

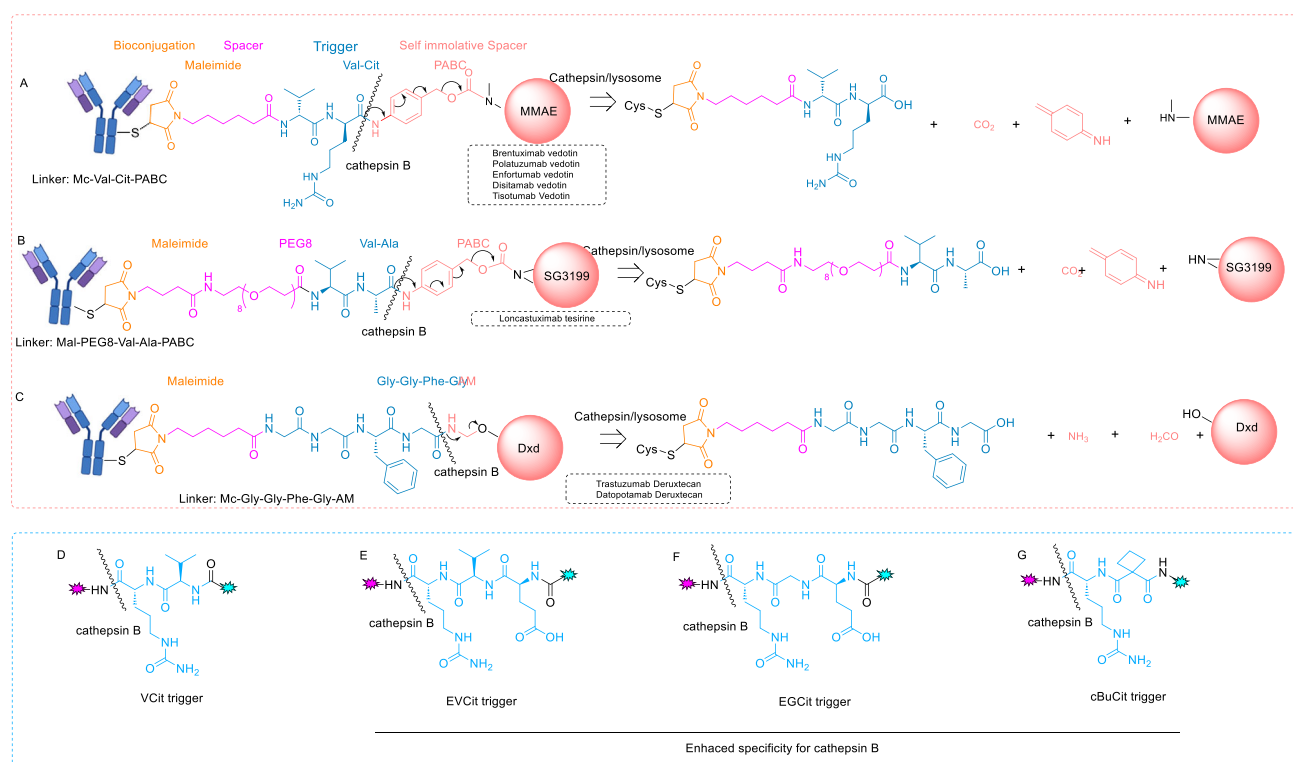
Of note, in addition to peptide-based triggers liable to lysosomal proteases, triggers responsive to other lysosomal

enzymes such as sulfatase,^[53] glycosidase,^[54] phosphatase,^[55,56] caspase,^[57] legumain,^[58] and non-lysosomal enzymes like fibroblast activation protein^[59] have emerged as potential alternatives, which may require extensive clinical verification for popularity.

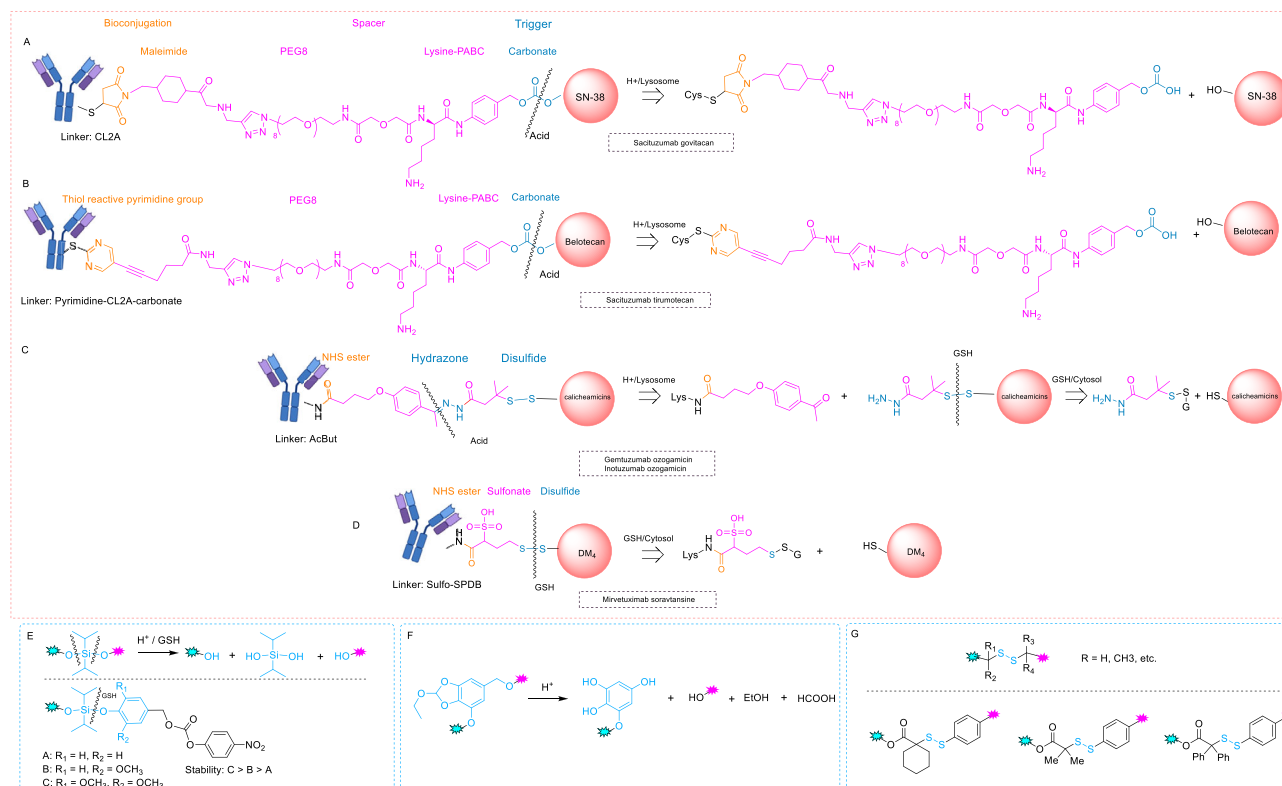
2.2. Acid-Labile Drug Release Trigger

Acid-labile triggers exploit the differences in pH between normal tissues, the tumor microenvironment (TME), intracellular endosomes, and lysosomes to enable selective drug release. Under physiological conditions, the interstitial pH of normal tissues and the bloodstream is stable at ≈ 7.3 – 7.4 .^[60] In contrast, the TME is generally more acidic due to increased glucose metabolism and poor perfusion, with pH levels dropping to as low as 5.6 in certain tumor types.^[61,62] Additionally, lysosomal pH ranges from 4.5 to 5.5, while endosomal pH in normal cells typically falls between 5.0 and 6.0.^[63] Acid-labile triggers incorporate functional groups that remain stable at neutral pH but undergo cleavage under acidic conditions, ensuring selective payload release. Common chemical motifs used in these linkers include hydrazones, *cis*-aconityl, and acetal/ketal linkages, which hydrolyze in acidic environments.^[64]

Four ADCs with such acid-labile triggers was approved (**Scheme 4A–C**): gemtuzumab ozogamicin,^[65] inotuzumab ozogamicin,^[5] sacituzumab govitecan,^[10] and sacituzumab tirumotecan.^[66] gemtuzumab ozogamicin consists of an anti-CD33 antibody conjugated to N-acetyl- γ -calicheamicin dimethylhydrazide via a



Scheme 3. Linker structure and drug release of enzyme liable ADC. A) Linker of 5 approved ADC: brentuximab vedotin, polatuzumab vedotin, enfortumab vedotin, disitamab vedotin, and tisotumab vedotin. B) Linker of loncastuximab tesirine. C) Linker of trastuzumab deruxtecan and datopotamab deruxtecan. D–G) VCit, D) trigger responsive to cathepsin B but instable in plasma, while triggers including E) EVCit, F) EGcit, and G) cBuCit have enhanced specificity for cathepsin B compared to VCit trigger.



Scheme 4. Linker structure and drug release of acid and GSH liable trigger. A) Linker and drug release of sacituzumab govitecan. B) Linker and drug release of sacituzumab tirumotecan. C) Linker and drug release of gemtuzumab ozogamicin and inotuzumab ozogamicin. D) Linker and drug release of mirvetuximab soravertansine. E) Acid/GSH liable trigger based on silky ether moiety. F) Acid liable trigger based on natural product gallic acid moiety. G) GSH liable trigger based on disulfide bond with variable steric hindrance.

bifunctional AcBut (4-(4-acetyl-phenoxy)-butyric acid) linker with a half-life is 72.4 ± 42.0 h (3 ± 1.75 days).^[67] This linker contains a carboxylic acid group activated by N-hydroxysuccinimide (NHS) to form an NHS ester for lysine bioconjugation, a hydrazone as an acid-labile trigger, and a dimethyl disulfide separated by an aromatic spacer. The hydrazone facilitates controlled payload release in acidic environments. The calicheamicin payload, a cytotoxic agent that causes double-strand DNA breaks, is connected to the linker through a disulfide bond further stabilized by two methyl groups and an acetylated amino sugar within the bicyclic core.^[65] Inotuzumab ozogamicin has the same hydrazone-based linker–payload system but is directed against CD22 with a long half-life 12.3 days.^[5,68] Sacituzumab govitecan comprises an anti-TROP2 antibody, the SN-38 payload, and a CL2A linker with half-life 11.7–18.9 h (0.48–0.78 days).^[10,69] The CL2A linker includes a thiol-reactive maleimide (Mal) group, a triazole moiety for assembly, a hydrophilic PEG8 spacer, and a lysine–PABC with a carbonate bond as the acid-labile trigger. Under acidic conditions, the benzyl carbonate bond hydrolyzes, ensuring efficient payload release. Similar to sacituzumab govitecan, sacituzumab tirumotecan employs a sulfonyl pyrimidine–CL2A–carbonate linker structure with the benzyl carbonate bond as the acid-labile trigger, and a sulfonyl pyrimidine moiety formed by irreversible cysteine bioconjugation using a 2-methylsulfonyl pyrimidine reagent.^[66,70] The half-life is 56.3 h (2.34 days).^[71]

Beyond hydrazones and benzyl carbonate bonds, other acid-labile motifs have been investigated, including *cis*-aconityl groups,^[72] orthoester,^[73] and β -propionate.^[74] In 2016, Berkman et al. reported an acid-labile linker incorporating a phosphoramidate trigger. This linker exhibited high water solubility, maintained good stability at physiological pH, and hydrolyzed rapidly in acidic environments.^[75] In 2019, Zhou et al. developed an ADC with a linker containing a silyl ether-based moiety as an acid-labile trigger, featuring an increased half-life of longer than 7 days (Scheme 4E).^[76] However, in 2025, Chen et al. questioned this finding, suggesting that the silyl ether trigger may be slightly more stable in acidic conditions than at neutral pH, although no experimental details were provided.^[77] In 2022, Taddei et al. developed a novel pH-sensitive drug release system utilizing a 1–6 self-immolative, traceless cross-linker derived from the natural product gallic acid (Scheme 4F).^[78] This structure remains stable for 24 h in buffers above pH 6.6 and in plasma, and it rapidly degrades under acidic conditions at pH 5.5.

2.3. Redox-Liable Drug Release Trigger

Redox-labile triggers are chemical motifs designed to release therapeutic agents under reductive or oxidative conditions intracellularly. Glutathione (GSH) provides a reductive condition, while reactive oxygen species (ROS), including superoxide anion ($\text{O}_2^{\cdot-}$), hydroxyl radical ($\text{HO}\cdot$), and hydrogen peroxide (H_2O_2), can provide

oxidative conditions.^[79] Driven by a higher metabolism rate, tumor cells tend to have higher concentrations of GSH and ROS. For GSH, the intracellular concentration is high and can reach 10 mM, while the extracellular concentration, including in the blood, is about 2–20 μM .^[80] At least a fourfold higher concentration of GSH levels in tumor tissue compared to normal tissue has been observed.^[81] For ROS, the level of H_2O_2 , one of the most common ROS, shows more than a tenfold increase in cancerous cells compared to normal cells.^[82] Additionally, chemotherapy can further increase ROS levels.^[83] With these differences between tumor cells and normal cells, redox-labile drug release triggers with disulfide (S–S) and diselenide (Se–Se) bonds were developed. Disulfide is GSH labile,^[84] while diselenide is both GSH and ROS labile.^[85,86]

As shown in Scheme 4C,D, there are 3 of the 17 approved ADCs utilizing the GSH-labile trigger based on disulfide bonds: gemtuzumab ozogamicin,^[65] inotuzumab ozogamicin,^[5] and mirvetuximab soravtansine.^[17] As mentioned earlier, gemtuzumab ozogamicin and inotuzumab ozogamicin have the same linker with a disulfide bond-based trigger in addition to the acid-labile trigger. Mirvetuximab soravtansine comprises a folate receptor α (FR α) directed antibody conjugated to DM4, a microtubule inhibitor, via a N-succinimidyl 4-(2-pyridyldithio)butyrate (sulfo-SPDB) linker. This linker includes a highly polar sulfonate group for enhanced solubility, a carboxylic acid group activated by NHS to form an NHS ester for lysine bioconjugation, and a thiol-protecting group capable of forming a disulfide bond with payloads or cysteine.^[17,87] The linker is designed to be stable in circulation, and the cleavage of the linker by GSH occurs in the cytosol instead of the lysosome.^[88] The half-life is 4.8–5.8 days.^[89]

Research on disulfide bonds for linker design mainly focuses on steric hindrance to increase plasma stability (Scheme 4G). Singh et al. in 2011 conducted research on steric hindrance around the disulfide bond and found varied drug release efficiency in ADCs with anti-CanAg and maytansinoid as the antibody and payload.^[90] Jiang et al. in 2020 reported a novel strategy for steric and stereoscopic disulfide construction.^[91] They created a synthetic method for diverse disulfides with varied steric hindrance using N-dithiophthalimides reagents, and 3 products were shown in the lower part of Scheme 4G. In addition to conventional disulfides, Chen et al. in 2025 found a silyl ether-based phenoxy silyl trigger structure with tunable stability due to its vulnerability to GSH (Scheme 4E).^[77] The drug release sensitivity in response to GSH can be tuned by changing the R_1 and R_2 groups attached to the phenyl ring. They tested the H_2O and methoxy group (OCH_3) for R_1/R_2 and found that GSH-mediated drug release was minimal with a GSH concentration at 1 mM. When the GSH concentration reached 10 mM, the drug release showed efficiency from rapid when $\text{R}_1/\text{R}_2 = \text{H}_2\text{O}$ to negligibly low when $\text{R}_1/\text{R}_2 = \text{OCH}_3$.

Like disulfide bonds, diselenide bonds break under the redox conditions of the TME. Lim et al. conducted research comparing diselenide- and disulfide-bond-containing linkers in core-cross-linked micelles in terms of stability and drug release.^[92] Under physiological conditions, S–S and Se–Se bonds remained stable. A concentration of 10 mM GSH triggered the de-cross-linking of these bonds, while 100 mM H_2O_2 only affected Se–Se. Drug release was lower at pH 7.4 and higher at tumorlike pH 5.

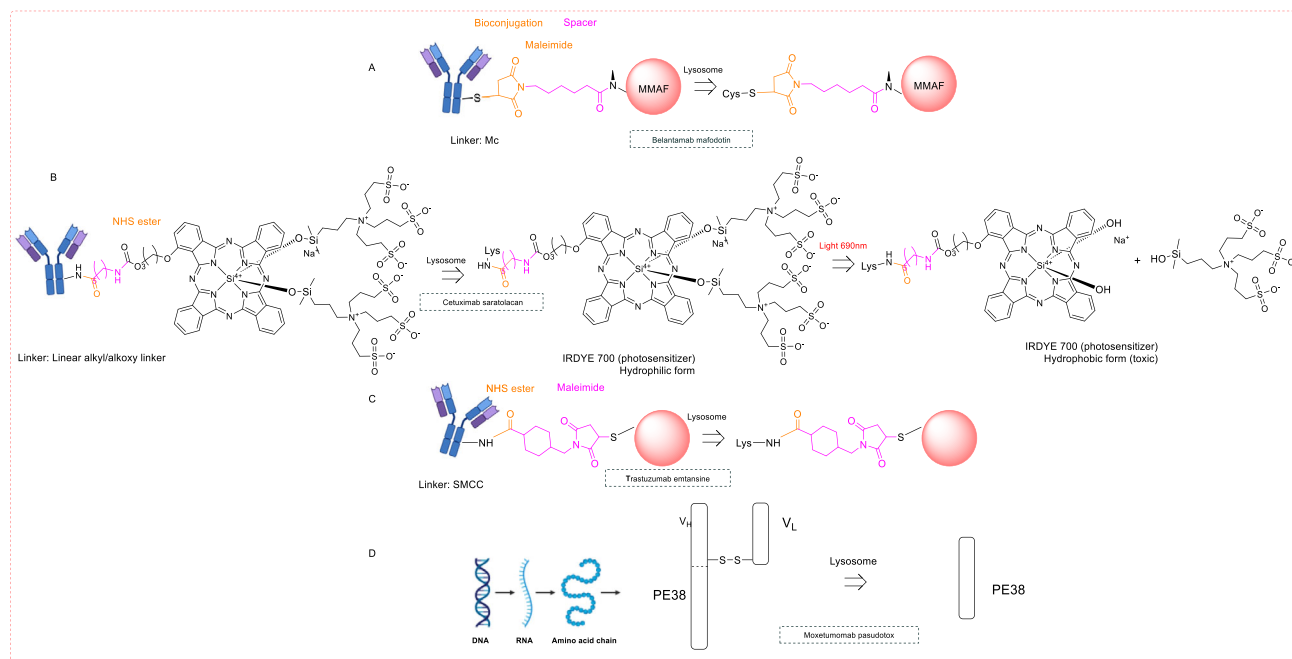
Dynamic light scattering techniques also found that micelles containing diselenide bonds are more sensitive to redox conditions compared to disulfide bonds, indicating higher efficiency for drug release. When loaded with doxorubicin, the micelles were safe for HEK-293 cells but effectively killed BT-20 cancer cells. To our knowledge, there have not been reports of ADC development with diselenide bonds, although their feature as a dual redox-labile motif holds noticeable value for ADC development. Although diselenide bonds seem easier to break due to their lower bond energy (172 kJ mol^{-1}) compared to disulfide bonds (240 kJ mol^{-1}),^[93] the diselenide bond of GSeSeG is 7 kcal mol^{-1} more stable than the disulfide bond of GSSG because its reduction potential ($\approx 400 \text{ mV}$) is about 151 mV lower than the disulfide bond.^[94]

2.4. Linker without Drug Release Trigger

Without drug release triggers, linkers are categorized as non-cleavable linkers that support a higher stability of the linker-payload structure in circulation with less premature drug release compared to linkers with drug release triggers.^[95] Such linkers remain intact in circulation until the entire ADC is internalized into the early endosome. The ADC is then transferred into the late endosome and lysosome. The drug release to the cytosol depends on the proteolysis of the antibody into individual AAs in the lysosome. Eventually, a single AA with its side chain connected to the linker-payload is released from the lysosome.^[31,96,97] The attached AA depends on the AA chosen for bioconjugation, such as cysteine or lysine (Scheme 5A–C).

There are 4 approved ADCs constructed without drug release triggers: belantamab mafodotin,^[11] cetuximab saratolacan,^[13] trastuzumab emtansine,^[25] and moxetumomab pasudotox.^[7,98] Inotuzumab ozogamicin is a humanized IgG1 ADC-targeting B-cell maturation antigen (BCMA) with the microtubule inhibitor MMAF as the payload and a long half-life 11.5 days.^[11,99,100] MMAF binds to tubulin and inhibits its polymerization, leading to G2/M phase arrest and subsequent apoptosis. The linker is a protease-resistant MC linker (mc) without a drug release trigger (Scheme 5A). Adotrastuzumab emtansine is an HER2-targeting ADC with the anti-mitotic agent mertansine (DM1) as the payload.^[25] The DM1 moiety inhibits tubulin polymerization, interferes with mitosis, and promotes apoptosis. The linker is named SMCC and lacks a drug release trigger (Scheme 5C). This linker features NHS ester and maleimide groups that allow covalent conjugation of amine- and sulfhydryl-containing molecules. The half-life is 3.5 days.^[101]

Cetuximab saratolacan is the first ADC approved for photoimmunotherapy. The anti-EGFR (epidermal growth factor receptor) antibody cetuximab is attached to the photosensitizer IRDye700DX via a linear alkyl/alkoxy linker (Scheme 5B). IR700 is a highly hydrophilic phthalocyanine compound with 2 water-soluble axial ligands. Once activated by near-infrared (NIR) light (690 nm), the two hydrophilic ligands are cleaved from the molecular backbone, converting the payload to an insoluble form. This change leads to the aggregation of conjugates in the membrane, causing physical damage to the



Scheme 5. Linker structure and drug release of ADC without trigger. A) Linker and drug release of belantamab mafodotin. B) Linker and drug release of cetuximab saratolacan. C) Linker and drug release of trastuzumab emtansine. D) Linker and drug release of moxetumomab pasudotox.

membrane, inducing inflow of water and ions, and subsequent necrotic/immunogenic cell death.^[102] Additionally, its internalization in lysosomes induces singlet oxygen-mediated ROS species, but the effects are weaker than membrane damage.^[103] The linker of cetuximab saratolacan is composed of 6-aminohexanoic acid with the carboxylic acid group turned into an NHS ester for bioconjugation with lysine on the antibody.^[103,104] Notably, Xu et al., in their review, classify cetuximab saratolacan as an ADC featuring an externally triggerable cleavable linker, with the drug being “released” in response to NIR light.^[105] However, since the released molecule from the ADC functions similarly to hydrophilic shields or the safety lever of a grenade, it’s debatable to categorize cetuximab saratolacan as having a cleavable linker. It seems more accurate to regard cetuximab saratolacan as an ADC with a non-cleavable linker and a photosensitive prodrug serving as the payload.

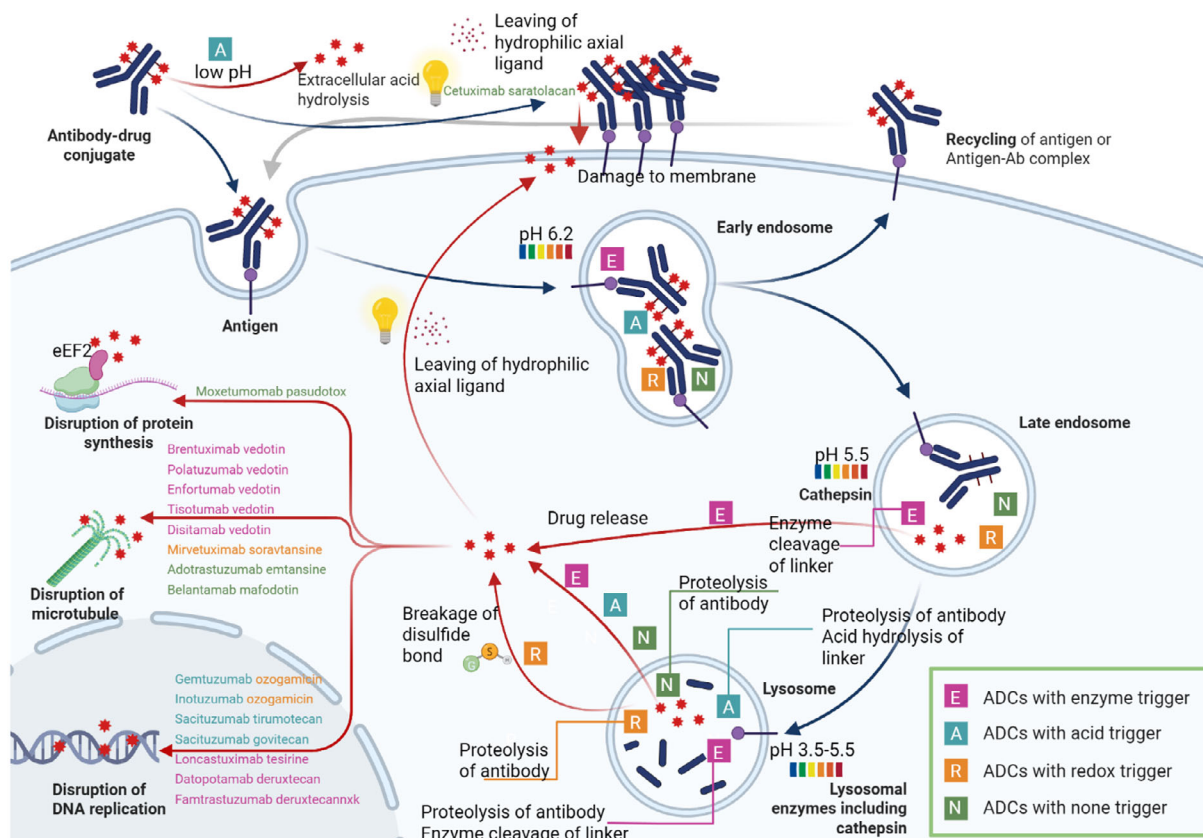
Moxetumomab pasudotox is an anti-CD22 recombinant immunotoxin comprised the toxin moiety PE38 genetically fused to an Fv fragment of an anti-CD22 monoclonal antibody.^[7] The drug was produced in *Escherichia coli* without chemical synthesis.^[106] PE38 is a 38 kDa fragment of *Pseudomonas* exotoxin A and can catalyze the ADP-ribosylation of the diphthamide residue in elongation factor-2 (EF-2), rapidly decreasing the level of the anti-apoptotic protein myeloid cell leukemia 1 (Mcl-1), inducing apoptotic cell death. Moxetumomab pasudotox contains a protein-based bacterial toxin as its payload and has a very short half-life of ≈ 1.4 h.^[107] This brief duration is at least partially attributed to its immunogenicity.

Although linkers without triggers are less popular, they are advantageous in systemic stability and off-target toxicity and are utilized by a variety of ADC pipelines developed in recent years. BDC-1001 is an immune-stimulating antibody–drug conjugate with a non-cleavable linker to minimize off-target immunotoxicity

caused by the TLR7/8 agonist.^[108] Although this drug had problems with low objective response rate (ORR) in phase 2, the well-tolerated profile is possibly attributed to the linker adopted. Xu et al. in 2021 reported BAT8001, an HER2-targeting ADC with a non-cleavable linker based on 6-maleimidocaproic acid and connected to the payload by a stable amide bond, which is the same as belantamab mafodotin (Scheme 5A).^[109] Jeong et al. in 2024 reported ABC101-1, a novel ADC-targeting HER2, with the microtubule inhibitor DM1 as the payload.^[110] This ADC utilized a non-cleavable linker based on maleimide, showing stronger tumor growth inhibition than trastuzumab emtansine with a DAR of 3.5 at the same dose. However, the detailed structure of the linker is not yet disclosed.

2.5. Trigger Influence Drug Release Pattern

Depending on the trigger type in the linker, ADCs have different patterns of drug release. Typically, ADCs undergo targeted internalization by binding to cell surface receptors. As shown in Scheme 6, the internalization is mediated by endocytosis. After that, the ADC molecules are first packaged in the early endosome, which matures into a late endosome with cathepsins capable of processing the ADCs with cathepsin triggers in the linker.^[111] Such ADCs include brentuximab vedotin, polatuzumab vedotin, enfortumab vedotin, trastuzumab deruxtecan, loncastumab tesirine, disitamab vedotin, tisotumab vedotin, and datopotamab deruxtecan. Later, the vesicles bearing the ADC molecules will merge with lysosomes where proteases are much more abundant. There are not only proteases like cathepsin recognizing specific peptide motifs but also proteolytic enzymes that nonspecifically digest the entire antibody. Through nonspecific proteolysis, the AA-linker–payload structure with one AA residue (usually cysteine



Scheme 6. This diagram illustrates the process of 17 approved antibody–drug conjugates (ADCs) from target binding to drug release, with a focus on distinguishing the drug release pattern by different triggering mechanisms (enzyme/acid/redox/non-cleavable). The symbols and color definitions in the figure are as follows: E (pink) indicates ADCs with lysosomal enzymes responsive drug release trigger, and their drug release primarily happens in late endosomes and lysosomes; A (light blue) indicates ADCs with acid responsive drug release trigger, and their drug release happens in lysosomes or in the acidic extracellular tumor microenvironment; R (orange) indicates ADCs with GSH responsive drug release trigger responsive to, and their drug release primarily happens in the cytosol. N (green) indicates ADCs without drug release trigger (non-cleavable linkers), and their drug release relies on proteolytic degradation of the entire antibody in lysosomes.

or lysine, depending on the bioconjugation method) is released. This process is important for the drug release of ADCs without a drug release trigger or non-cleavable linkers.^[97] These ADCs include trastuzumab emtansine, belantamab mafodotin, cetuximab saratolacan, and moxetumomab pasudotox. Notably, moxetumomab pasudotox is a full protein drug, while the payload (PE38) has been reported to be resistant to proteolysis.^[98] ADCs with acid-labile triggers undergo acid hydrolysis mainly in the lysosome due to the ultralow pH. These ADCs include gemtuzumab ozogamicin, inotuzumab ozogamicin, sacituzumab govitecan, and sacituzumab tirumotecan. For ADCs with redox-labile triggers, the cleavage of disulfide bonds mainly occurs in the cytosol, while breakage of diselenide bonds typically happens in the lysosome due to the biased concentration of GSH and ROS.^[92] ADCs with GSH triggers include gemtuzumab ozogamicin, inotuzumab ozogamicin, and mirvetuximab soravtansine.

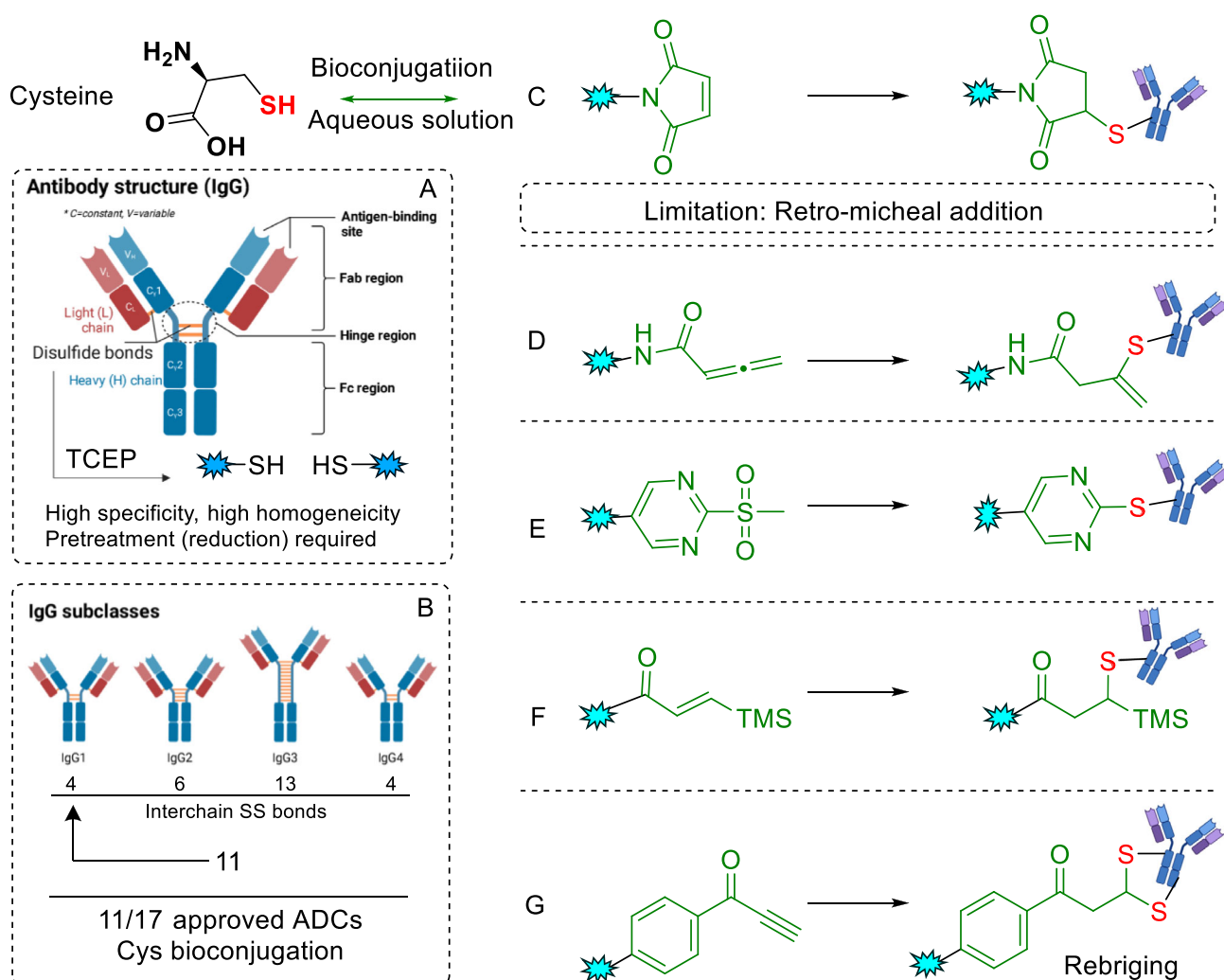
3. Bioconjugation for Attachment to Antibody

In ADC manufacturing, bioconjugation refers to the reaction that creates a linkage between the linker or linker–payload and the

functional groups of AA residues in the antibody. Cysteine, with its thiol group, and lysine, with its amine group, are the two most commonly targeted sites for bioconjugation.

3.1. Bioconjugation with Cysteine

Cysteine conjugation relies on the thiol groups present in the reduced form of cysteine residues in antibodies. Typically, mild reducing agents like Tris (2-carboxyethyl) phosphine (TCEP) or Dithiothreitol (DTT) are used to partially reduce interchain disulfide bonds, revealing reactive thiol sites (Scheme 7A). Since there are only four interchain disulfide bonds in IgG1 and IgG4, cysteine conjugation is more advantageous over lysine conjugation as it tends to yield more homogeneous ADCs that are more favorable for quality control. There are 15 approved ADCs with the IgG1 antibody bearing four interchain disulfide bonds, and 11 of them utilized the cysteine-based bioconjugation (Scheme 7B): brentuximab vedotin,^[3] polatuzumab vedotin,^[33] enfortumab vedotin,^[34] trastuzumab deruxtecan,^[9] sacituzumab govitecan,^[10] belantamab mafodotin,^[11] loncastuximab tesirine,^[112] disitamab vedotin,^[15] and tisotumab vedotin.^[16] All of them chose the maleimide-based conjugation except the newly approved Jatailai, which used a



Scheme 7. Common bioconjugation method with cysteine. A) Antibody structure and the reduction of disulfide bond into free thiol group. B) IgG antibody subclass. There are 11 of approved ADCs with IgG1 antibodies and cysteine dependent bioconjugation. Bioconjugation to cysteine with C) maleimide, D) allenamide, E) 2-methylsulfonyl pyrimidine reagents, F) silicon-containing thiol-specific reagent, and G) phenylpropynones based reagents for disulfide rebridging. D,F,G) These bioconjugation methods have not yet been used in approved ADC drugs; E) is involved in manufacturing of sacituzumab tirumotecan. Created with BioRender.

2-methylsulfonyl pyrimidine reagent (Scheme 4B). Of note, trastuzumab emtansine with lysine bioconjugation has the thiosuccinimide linkage between the spacer and the payload (Scheme 5C).

Maleimides are the most frequently used reagents in cysteine-dependent bioconjugation due to their high reactivity with thiols, forming stable thioether bonds (Scheme 7C). However, the thiosuccinimide linkage formed by maleimide-mediated bioconjugation with the thiol group of cysteine is unstable because of the retro-Michael reaction, causing the loss of the linker–payload from the antibody.^[113] Senter et al. in 2014 added a basic amino group adjacent to the maleimide by using diamionopropionic acid (DPR), positioned to provide intramolecular catalysis of the hydrolysis of the thiosuccinimide ring, converting the linkage to a form resistant to maleimide elimination.^[28] The stability test revealed that the ADC (anti-CD30 conjugated with MMAE) with the modified linker remained stable with little loss for 7 days, while the one without the modification lost about half of the payload in three days. The linker used was denoted

as self-stabilizing maleimido–DPR. Dimasi et al. in 2015 reported that ADC with bioconjugation by N-aryl maleimides has a lower deconjugation rate less than 20% compared to that of 35%–67% by N-alkyl maleimides,^[114] and the method was further utilized in work by Gao et al. in 2017 to produce a PBD ADC.^[115]

In addition to strategies making the thiosuccinimide linkage more stable, Abbas et al. in 2014 developed a method using allenamide, an alternative to maleimide for cysteine-based bioconjugation, during a study in Professor Teck Peng Loh's lab at NTU (Scheme 7D).^[116] The reaction between allenamide and the cysteine thiol group is swift in an aqueous buffer. They tested the peptide conjugates by incubation with excessive GSH/DTT and observed good irreversibility with the possibility for protein modification under biological conditions. Baud et al. reported research using 2-sulfonylpyrimidine reagents for cysteine bioconjugation under mild conditions in 2023 (Scheme 7E).^[70] This reagent is of high thermoelectricity toward the thiol group. In 2024, the Loh group reported another silicon-containing

thiol-specific reagent with high chemoselectivity and resistance to the retro process observed in the thiosuccinimide linkage by maleimide (Scheme 7F).

Despite the advantage of cysteine conjugation over lysine conjugation, there is still the problem of heterogeneity making quality consistency challenging. As such, disulfide rebridging strategies based on the four interchain disulfide bonds, with the potential for higher homogeneity, was developed. The disulfide rebridging method reduces the four interchain disulfide bonds and reacts with a cysteine-selective cross-linker. The reagent reconnects polypeptide chains and installs drug molecules or functionalities for further modification, maintaining the stability of disulfide bonds, allowing one linker per disulfide, and achieving a desired DAR of up to 4.^[117] Recent advances include methods with the 3,5-dibromo-1-(but-3-ynyl)pyrazin-1-ium triflate (BUPY), four-divinylpyrimidine (tetraDVP), and STEF reagents with higher oxidation state at C1 (oxSTEF). Keserü et al. recently produced the ADC via site-selective cysteine-based bioconjugation with dibromopyrazine.^[118] They synthesized a disulfide rebridging agent, BUPY. BUPY can react with thiols by reducing the four interchain disulfides of the antibody, providing four alkyne groups. The alkyne groups can then be used for conjugation via azide-alkyne cycloadditions to form the ADC. The Spring group conducted research using a tetraDVP moiety with an azide group to achieve disulfide rebridging and reported the technique to prepare single-payload ADCs using the tetraDVP linker with PEG modification. Poulsen et al. also reported a novel class of electrophilic reagents (oxSTEF) for selective disulfide rebridging in protein modification.^[119] Winne et al., in 2024, reported phenylpropynones as a novel disulfide rebridging reagent with high chemoselectivity for cysteine over the amine group of lysine (Scheme 7G).^[120]

3.2. Bioconjugation with Lysine

Lysine conjugation is one of the earliest and most straightforward methods due to the abundance of lysine residues (≈ 80 sites in IgG antibodies) on antibodies, with potential for higher DAR value (Scheme 8A).^[121] Each lysine has an ϵ -amino group that can react with activated esters or isocyanates to form stable amide or urea linkages.^[122] Unlike cysteine conjugation, which requires the reduction of disulfide bonds and may potentially perturb the natural structure and function of the antibody, lysine-based bioconjugation does not require pretreatment that could perturb antibody structure. There have been five approved ADCs produced by lysine-dependent bioconjugation (Scheme 8B): gemtuzumab ozogamicin,^[2] trastuzumab emtansine,^[123] inotuzumab ozogamicin,^[5] cetuximab saratolacan,^[13] and mirvetuximab soravtansine.^[17]

For bioconjugation with lysine, the most popular reagent is NHS ester, which was first developed by Callahan et al.^[124] NHS esters react efficiently with the primary amines in lysine side chains to form stable amide bonds, effectively modifying biomolecules such as antibodies or enzymes (Scheme 8C). This technique benefits from being relatively fast and proceeds under mild conditions (pH 7–9), preserving protein function. NHS esters

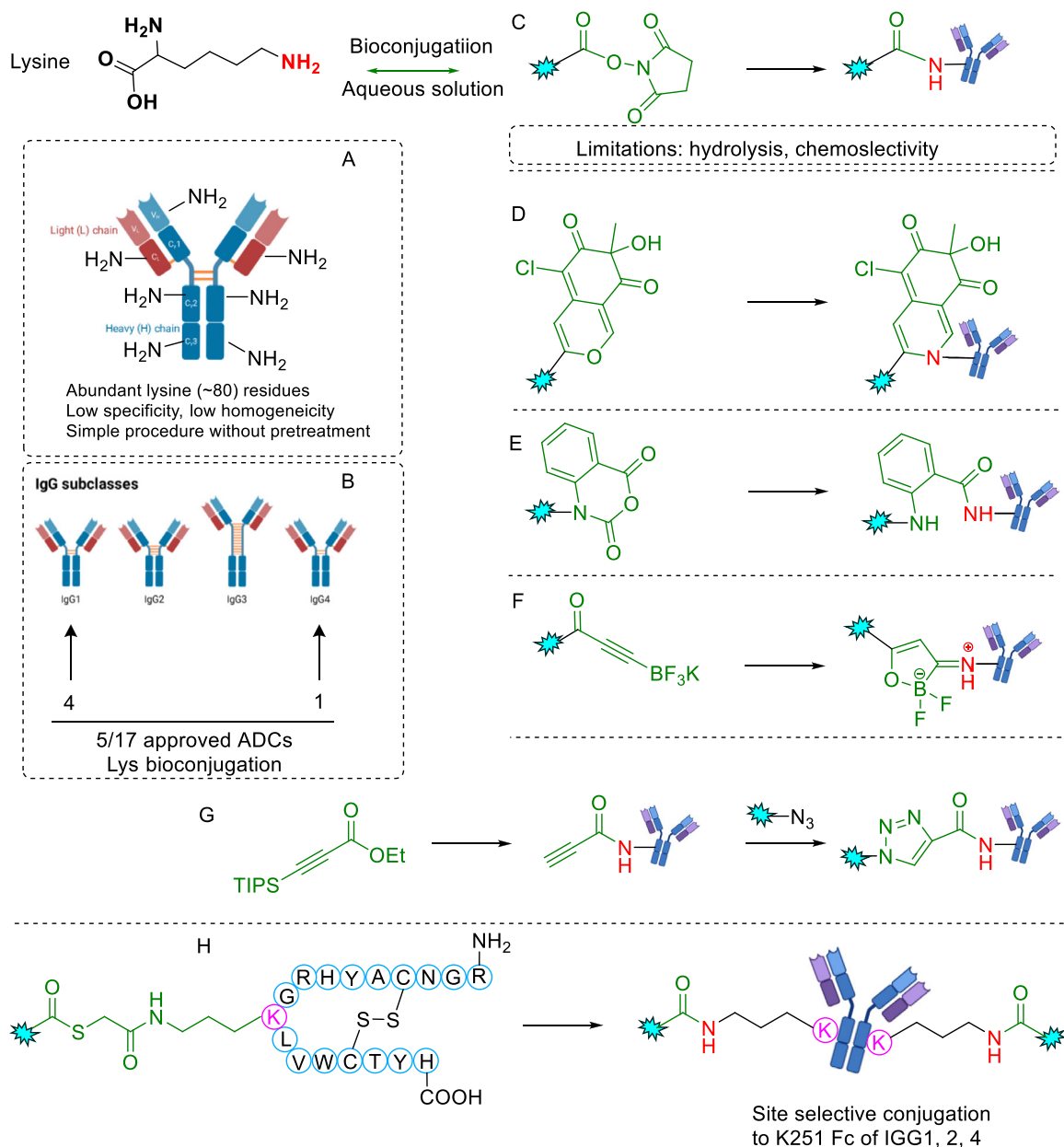
were adopted for producing the first generation of ADCs, including approved trastuzumab emtansine,^[123] gemtuzumab ozogamicin, and inotuzumab ozogamicin.^[125] However, NHS ester-mediated bioconjugation with lysine lacks selectivity, as multiple accessible lysine residues can lead to heterogeneous products.^[126] Susceptibility to hydrolysis and picky pH requirements (typically 7.2–8.0) restrict the application in ADC manufacture.

Consequently, a variety of novel reagents have been developed. The Yao group reported an azaphilone-based reagent for bioconjugation with the primary amine of biomolecules in 2021 (Scheme 8D).^[127] This reagent showed high chemoselectivity for the amino residue of lysine side chains in proteins under mild conditions, with only one water molecule as the byproduct. Additionally, the bond formed was tested in human serum and showed good stability for up to 7 days. In 2024, Teng et al. (Loh group) developed novel amine-specific biocompatible click reagents using alkynone β -trifluoroborates as the electrophiles (Scheme 8F).^[128] In competition studies between benzylamine and benzyl mercaptan, the alkynone β -trifluoroborate reagent showed high chemoselectivity for amine addition (82% yield) over thiol addition (trace) under mild conditions. The boron-containing alkynyl reagent demonstrated significant potential for lysine-based bioconjugation, with great potential for the development of ADCs. Additionally, the same group reported another method to form the amide bond with potential for bioconjugation (Scheme 8G). Allimuthu et al., in 2024, discovered isotonic anhydrides as a lysine-selective bioconjugation reagent (Scheme 8E).^[129]

Despite the advantage of lysine conjugation that does not require pretreatment, this method leads to heterogeneous ADCs, as the reaction occurs randomly across multiple lysines, producing a mixture of different species. It was reported that adotrastuzumab emtansine contains 88 lysines and 4 N-terminal groups susceptible to lysine bioconjugation, and about 70 of these sites are involved in the bioconjugation (Scheme 8A).^[130,131] Such heterogeneity can affect the ADC's stability and binding affinity, posing challenges in manufacturing consistency and regulatory approval. Thus, site-specific bioconjugation using affinitive peptide-guided, proximity-driven group transfer reactions with certain lysine residues (frequently K248) has emerged as a method to ease the manufacture of high-quality ADCs. The Huang and Tang group, in 2022, reported a protocol for site-specific bioconjugation of lysine (K248) in IgG antibodies (Scheme 8H).^[132] They engineered an Fc fragment-binding peptide (FcBP) attaching the linker–payload structure for optimal bioconjugation efficiency. Xia et al., in 2022, reported another protocol to introduce an azide group into the side chain of lysine 248 of the heavy chain of human immunoglobulin G (IgG) using a phenolic ester as the activated acetyl group donor and an optimized FcBP mimicking the lysine acetyltransferase framework.^[133]

3.3. Emerged Bioconjugation Strategies

Conventional methods based on random bioconjugation with cysteine or lysine have been thoroughly verified by the 17 approved ADCs. Nonetheless, the manufacturing of ADCs with conventional



Scheme 8. Common bioconjugation methods with lysine. A) Feature of lysine bioconjugation. B) IgG antibody subclass. There are 5 approved ADCs (4 IgG1 and 1 IgG4) with lysine bioconjugation. Bioconjugation to lysine with C) NHS ester, D) Azaphilones, E) Isotonic anhydrides, F) Alkynone β -trifluoroborates, G) β -Silyl alkynoates, and H) FcBP-guided group transfer reagents. D–H) These bioconjugation methods have not yet been used in approved ADC drugs. Created with BioRender.

bioconjugation methods is challenging for achieving consistent DAR across batches. Consequently, strategies have been developed to overcome this problem through site-specific conjugation, such as non-natural AA (NAA) incorporation, Thiomab, enzyme-dependent technologies like GlycoConnect, etc. NAA-based technology involves the genetic insertion of an NAA with a specific functional group serving as a unique conjugation site in the antibodies, such as *p*-acetophenylalanine (pAcF) carrying a ketone group and *p*-azidophenylalanine (pAzF) carrying an azide group in the side chain.^[134] Although the NAA-based method may have issues in manufacturing, mainly due to the incorporation efficiency of the NAA and potential immunogenicity, it guarantees

high consistency of the DAR across batches. Thiomab technology by Genentech involves the incorporation of a new cysteine by mutagenesis of the antibody to serve as a unique conjugation site, enabling high consistency. GlycoConnect is a technology by Synaffix that uses an engineered endoglycosidase and a native glycosyltransferase to attach a sugar substrate, 6-azidoGalNAc, into the globally conserved N-glycosylation site of the antibody.^[135] With copper-free click chemistry, the linker-payload structure can be attached to the antibody with high consistency. Delft et al. from Synaffix in 2015 reported the GlycoConnect technique for ADC development.^[136] This technique utilizes endoglycosidase to trim the glycosylated Asn297 of antibody, and then tag a

azide group with galactosyl transferase. The subsequent azide group can serve as a site to attach the linker–payload with copper-free click chemistry method, preferentially by reagent BCN for the click chemistry. This technique doesn't require genetic engineering of the antibody and suitable for all IgG monoclonal antibodies. With this advantage, it has been utilized by recent ADC pipelines including IBI343, IBI3001 by Innovent, MGC026, MGC028 by MacroGenics, XMT-1660 by Mersana Therapeutics, ADCT-601, and ADCT-701 by ADC therapeutics.^[137]

4. Spacer

Once the bioconjugation method and drug release trigger have been established, the design of spacers becomes the next focus. Spacers act as connectors linking essential core elements such as the bioconjugation group, drug release trigger, and payload. With technological advancements, spacers have emerged as a key area of interest in linker design because they impact hydrophilicity, drug release, and the branching of linker.

4.1. Spacer for Enhanced Hydrophilicity

Most payloads in ADCs are highly hydrophobic, including popular compounds such as MMAE and exatecan. Consequently, a major challenge in ADC manufacturing is the hydrophobicity caused by the linker–payload structure, which can induce aggregation and lead to an unfavorable Pharmacokinetics (PK),^[138] posing challenges in manufacturing and storage.^[126,139] To address this issue, a common strategy is the incorporation of hydrophilic spacers such as PEG8, **Scheme 9A**), as used in sacituzumab govitecan and sacituzumab tirumotecan, or polar groups like sulfamate (Scheme 9B) seen in mirvetuximab soravtansine that reviewed by Sander et al in 2021.^[140] PEG, comprising repeating units of ethylene oxide, is extensively utilized in linker design for its safety, stability, and solubility.^[141] In addition to PEG and sulfamate, other spacer technologies being researched for ADC linker design include

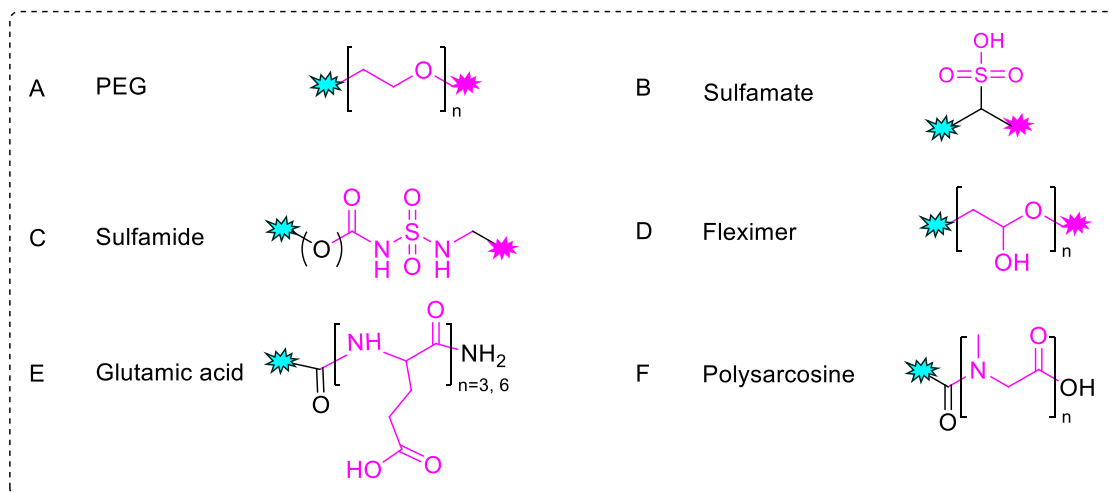
sulfamide (Scheme 9C), glutamic acid (Scheme 9E), fleximer (Scheme 9D), and polysarcosine (Scheme 9F).

In 2018, Delft et al. from Synaffix reported a polar acyl/carbamoyl sulfamide spacer (HydraSpace) and its application in ADC development.^[142] In a 3 h test, they observed that incorporating this spacer significantly enhanced conjugation efficiency to nearly 100%, compared to $\approx 20\%$ when using only the PEG spacer. This technology has been applied in multiple ADC developments in recent years, including IBI343, an anti-claudin18.2 antibody, and exatecan conjugate initiated by Innovent (phase II/III), as well as MRG004A, an anti-TF antibody, and MMAE conjugate initiated by Lepu Biopharma (phase I/II).

In 2019, Orlova et al. reported on research testing hydrophilic spacers using glutamic acid in an HER2-targeting ADC with the cytotoxic maytansine derivative MC-DM1 as the payload.^[138] They found that incorporating the hydrophilic spacer significantly reduced the hepatic uptake of the ADC from $13.4\% \pm 0.9\%$ ID g^{-1} to $8.7\% \pm 0.2\%$ (triglutamate–spacer) and $8.6\% \pm 0.9\%$ (hexaglutamate–spacer). In 2021, Lowinger et al. from Mersana Therapeutics reported an ADC platform featuring high drug loading with a hydrophilic spacer composed of poly-1-hydroxymethylethylene hydroxymethylformal (fleximer).^[143] This spacer enables a DAR of up to 10–15, with favorable pharmacokinetic properties and antitumor potency. Polysarcosine was another hydrophilic spacer extensively researched for ADC linker design.^[144]

4.2. Spacer for Traceless Drug Release

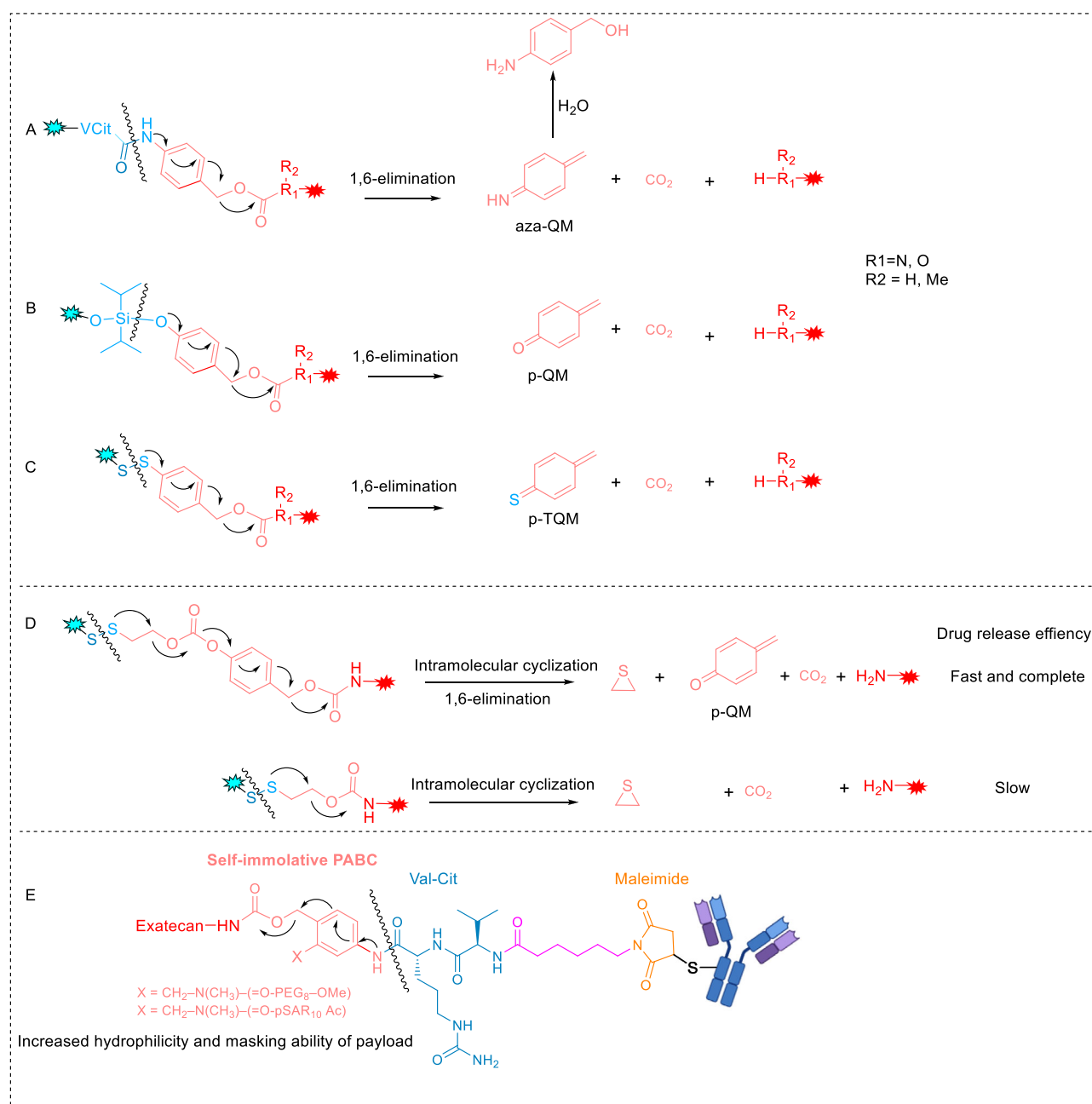
Self-immolative spacers play an important role in the drug release of ADCs with cleavable linkers. These spacers act as dynamic chemical bridges between the drug-release trigger and the payload and are engineered to undergo controlled, multi-step disassembly in response to triggers activated by desired conditions such as lysosomal proteases, acidic pH, or reducing environments.^[145] The unique “domino-like” degradation mechanism ensures that the payload is liberated in its native, unmodified form without leaving behind residues that might impair drug



Scheme 9. Examples of spacers to enhance the hydrophilicity of linker. A) PEG, B) sulfamate, C) sulfamide, D) fleximer, E) glutamic acid, and F) polysarcosine.

activity or induce unintended immune responses.^[146] To achieve traceless drug release, three functional segments are needed: the trigger unit, the self-immolative chain, and the attachment to the drug.^[146] In ADC linkers, the trigger unit is the drug-release trigger. The self-immolative spacer is a chemical structure that undergoes spontaneous, sequential fragmentation after the trigger is activated. The attachment to the payload is a covalent bond designed to detach cleanly upon the breakdown of the spacer. The PABC spacer, based on quinone methide (QM) species, is perhaps the most popular one.^[147] PABC and its byproducts have been validated in five approved ADCs (Table 1). PABC is frequently

triggered by enzyme-labile triggers, such as the VCit trigger. The attachment of PABC to the payload is through carbamate or carbonate bonds designed to detach cleanly when PABC breaks down. Upon ADC internalization into cancer cells, the drug-release trigger is activated in the lysosome or cytosol under conditions like proteases. This initiates a cascade of 1,6-elimination of the PABC spacer, culminating in the release of the unmodified drug, aza-QM, and carbon dioxide (**Scheme 10A**). In addition to peptide-based triggers, self-immolative spacers based on QM have also been studied with other triggers.^[77] For example, they have been studied with silyl ether-based triggers as mentioned earlier (**Scheme 10B**)^[77] and



Scheme 10. Examples of self-immolative spacer. A) Drug release mechanism of self-immolative PBAC spacer. B,C) Example of other spacer based on QM species. The drug release depends on disulfide or silyl ether-based trigger in response to reduction. D) A novel self-immolative spacer with improved performance on drug release efficiency. E) PABC spacer modified for improved hydrophilicity to mask the hydrophobic payloads.

with disulfide-based triggers as summarized by Liu et al. in 2020 (Scheme 10C).^[148]

One research focus of self-immolative spacers is on drug release efficiency. In 2021, Zhang et al. reported a self-immolative spacer triggered by the breakdown of the disulfide bond in response to reduction by GSH.^[149] Compared with spacer elimination by intramolecular cyclization, the novel spacer combined the elimination mechanisms of intramolecular cyclization and 1,6-elimination of QM together to achieve fast and complete drug release (Scheme 10D). This research provides the possibility of improvement for antibody–PBD conjugates with disulfide-based triggers and self-immolative spacers eliminated by intramolecular cyclization, as reported by Pillow et al. in 2017.^[150] In 2024, Corso et al. reported an amine-carbamate-based self-immolative spacer releasing tertiary alcohol at faster rates than primary alcohol payloads like R848.^[151] This research would be useful for developing immune stimulation ADCs with R848 as payloads.

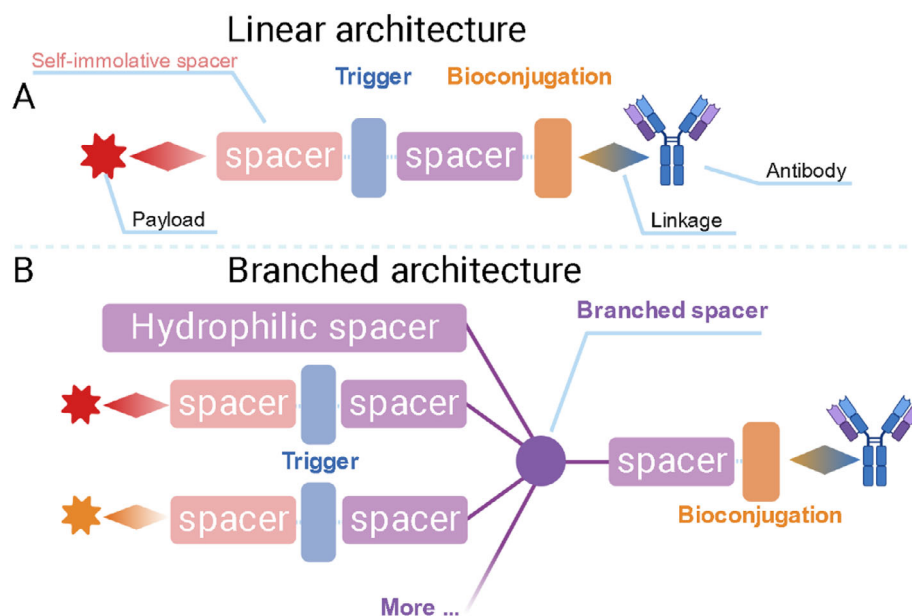
Another research focus is on modification for hydrophilicity. In 2023, Meng et al. designed a novel self-immolative spacer for antibody–exatecan conjugates to overcome resistance in colon and lung cancer.^[152] The spacer, which is based on PABC, was modified with either PEG8 or pSAR10 (Scheme 10E). The PEG8 modification resulted in an ADC aggregation level of 2%, which is considered quite acceptable. Further modification with pSAR10 reduced the aggregation level to a negligible amount. In vivo results demonstrated that this ADC modification achieved a durable response in low-target-expression tumors resistant to Dxd/SN-38 ADCs without increasing toxicity. This modified spacer effectively enhances hydrophilicity, helping to mask the hydrophobic payload and improve manufacturing quality without additional efforts on the linker design.

4.3. Spacer for Branched Linker

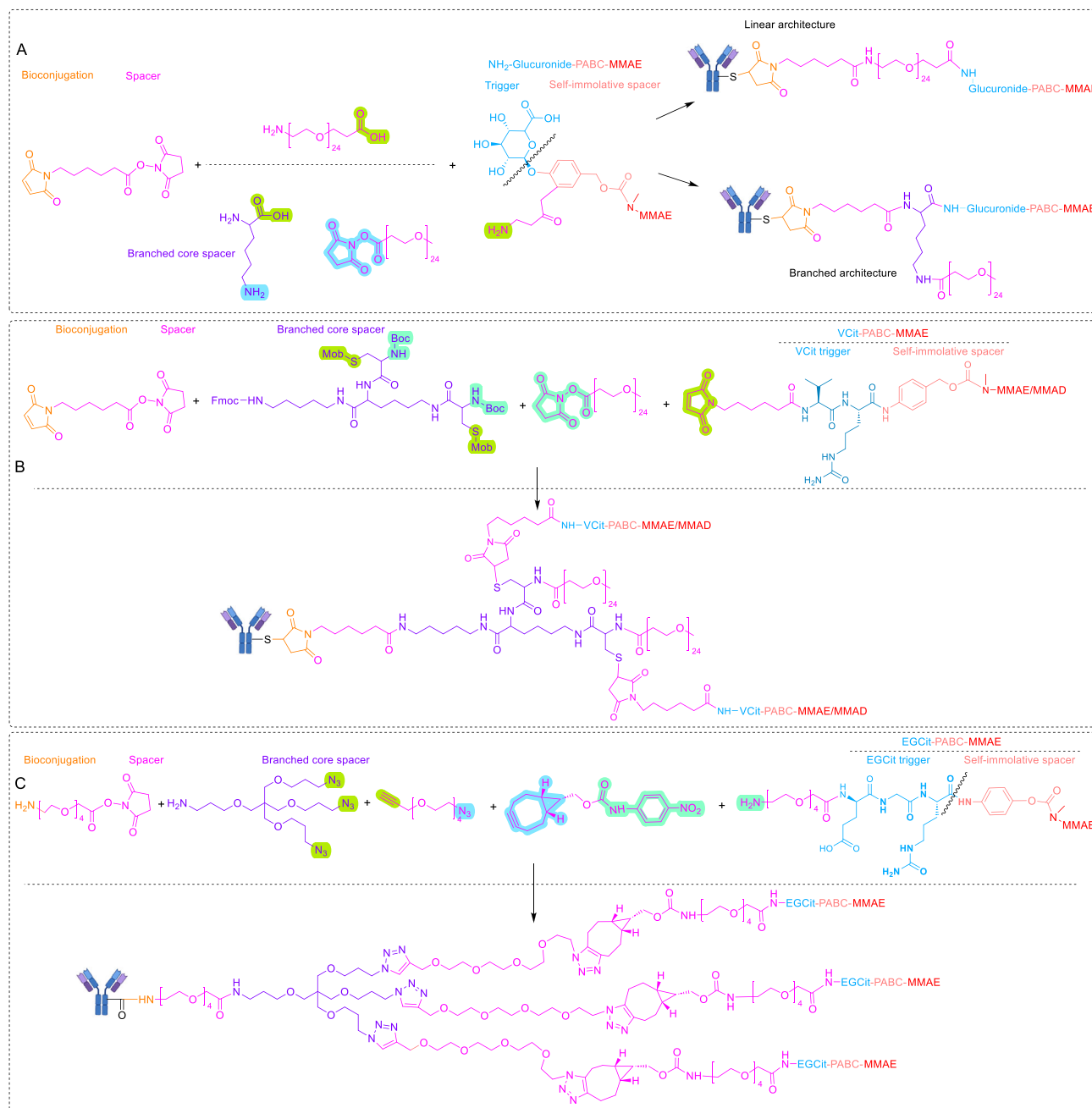
Branched linkers, designed with branches for hydrophilic moieties (e.g., PEG) and/or the attachment of multiple payloads on a single-linker scaffold with a branched spacer, provide significant advantages over traditional linear linkers. These benefits include enhanced hydrophilicity, greater drug loading capacity, and the ability to accommodate versatile drug combinations. Structurally, branched linkers are synthesized by incorporating a branched spacer as the assembly core (Scheme 11).

In 2015, Senter et al. studied the differences between a linear linker and a branched linker with lysine as the branched core (Scheme 12A).^[153] They explored how the placement of the PEG24 moiety influences ADC drugability and found that the ADC with PEG24 in the side branch demonstrated higher performance regarding pharmacokinetics and therapeutic index compared to the linear linker. According to plasma pharmacokinetics data in rats, the branched linker achieved high performance in terms of elimination, whereas the linear linker with the PEG24 moiety performed worse than the linker without PEG24 due to increased overall hydrophobicity of the ADC, attributed to the exposure of the hydrophobic surface of the payload.

In 2018, Dhal et al. reported a branched linker designed with a spacer including a branched spacer core based on a lysine and two cysteines (Scheme 12B).^[154] With the carboxylic group of lysine connecting the maleimide for bioconjugation, the remaining two amino groups served as sites for conjugation with the two cysteines through an amide bond. Subsequently, the thiol group was loaded with the trigger–spacer–payload (VCit–PABC–MMAE/MMAD), and the amino group was loaded with a PEG24 to provide adequate hydrophilicity and reduce aggregation. Using this linker, they developed an HER2-directed ADC with a DAR of 5.6, which



Scheme 11. Schematic comparison of linear architecture and branched architecture of ADC linker. A) Linear linker with typical bioconjugation group, spacer, drug release trigger, and self-immolative spacer for attachment to payload. B) Branched linker featured with a branched spacer as the assembly core to connect different components of the linker. With such architecture, advantage like multiple payloads, combination of different payloads as well as higher hydrophilicity without risk increasing hydrophobic exposure of the payloads would be possible. Created in BioRender.



Scheme 12. Examples of branched linker. A) Branched linker with lysine as the branched core. B) Branched linker with a peptide based branched spacer comprised one lysine and two cysteines. C) Branched linker with amino-triazide as the branched core spacer for assembly.

remained highly monomeric with aggregation less than 3%. In contrast, the ADC with a conventional linear linker and a DAR of 3.7 showed about 44% aggregation. They further increased the DAR to 10.8 using the branched linker and observed only about 22% aggregation. Notably, they found that the hydrophilicity provided by PEG24 was crucial for this architecture.

In 2021, Tsuchikama et al. reported an HER2-directed ADC with a branched linker using two lysines as the branched spacer core.^[155] Using the two lysines, they loaded both MMAE and MMAF onto the ADC in ratios of 1:1, 2:1, and 1:2. This strategy enables a variety of combinations of different payloads for maximized antitumor effects.

In 2024, Sapozhnikova et al. developed a novel branched linker with three payloads attached for ADC development (Scheme 12C).^[156] They first synthesized a branched spacer based on amino-triazide as the core for linker assembly. The amino group was used to form an amide bond with the antibody via enzyme-catalyzed site-specific conjugation, and the three-azide groups were used to attach the trigger-spacer-payload moiety (EGCIt-PABC-MMAE). To support this architecture, multiple PEG molecules were incorporated linearly into the individual branches (Scheme 12C). Between the branched spacer and the trigger-spacer-payload moiety, they inserted PEG4 to increase hydrophilicity and studied whether this incorporation was

necessary. Interestingly, the branched linker with more PEG4 was more hydrophilic but made the ADC slightly more hydrophobic, possibly due to increased exposure of the payload's hydrophobic surface. Nevertheless, the ADC with longer PEG achieved much higher efficacy in *in vitro* cytotoxicity compared to the ADC without PEG4 insertion. The difference was explained by impaired efficiency in drug release due to steric hindrance to enzyme cleavage.

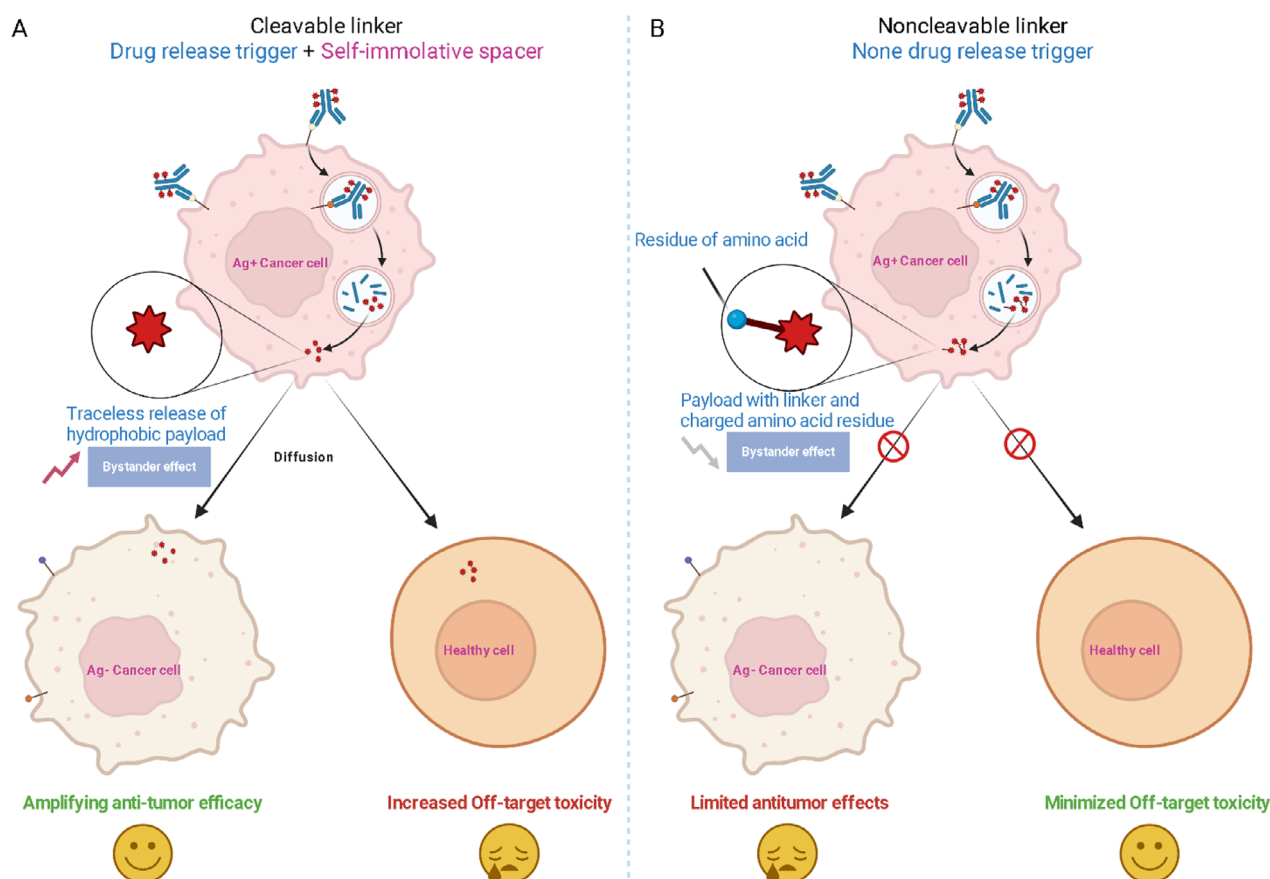
In addition to higher payload capacity, the branched linker enables a combinatorial strategy of different payloads. In summary, branched linkers with branched spacers represent a transformative advancement in ADC technology, addressing key limitations of traditional linkers. Through improvements in hydrophilicity, drug-loading capacity, and combinatorial payload strategies, they pave the way for more potent ADCs with broader therapeutic applications in oncology.

5. Consideration of Bystander Effect for Linker Design

The bystander effects of ADCs refer to the ability of payloads released from target cells to diffuse to neighboring cells, causing cell death.^[157] Since there is no antibody-directed specificity, this

diffusion may damage neighboring cancer cells as well as healthy cells, either amplifying antitumor efficacy or exacerbating off-target toxicity (Scheme 13).^[157,158] This phenomenon depends on payload permeability and the biochemical properties of the linker, which govern payload release dynamics and spatial distribution.^[159] Cleavable linkers can achieve traceless drug release, preferably triggered by a specific factor, and a self-immolative spacer promotes the bystander killing of neighboring tumor cells, increasing antitumor effects. This approach is favorable for targeting antigen-negative cancer cells (Scheme 13A). In contrast, non-cleavable linkers lead to drug release with the linker attached, often leaving a residue of charged AA. This is not conducive to drug molecule diffusion needed for bystander effects but helps minimize off-target toxicity to healthy cells (Scheme 13B).

For example, with an enzyme-labile trigger and a self-immolative spacer attached to the membrane-permeable payload (Dxd), trastuzumab deruxtecan (DS-8201a) demonstrated bystander effects resulting in potent antitumor efficacy in a mouse xenograft model with heterogeneous HER2 expression.^[159] In contrast, ado-trastuzumab emtansine, lacking a drug release trigger, was less effective against HER2-negative cells in tumor tissues.^[159] Due to this capability, trastuzumab deruxtecan has been approved for HER2-low breast cancer.^[160] However, bystander effects have been implicated in lung injury during treatment.^[161,162]



Scheme 13. Impact of linker type on bystander effect: A) cleavable linkers enhance the bystander effect, boosting antitumor activity against antigen-negative cancer cells, though they may increase the risk of off-target toxicity to healthy cells. B) Non-cleavable linkers are less conducive to drug diffusion necessary for bystander effects but effectively reduce off-target toxicity to healthy cells. Created in BioRender.

Beyond the differential influence of linker types on bystander effects, acid-labile triggers may offer advantages over enzyme-labile triggers. Thurber et al. (2023) studied sacituzumab govitecan (hRS7-CL2A-SN-38) with a CL2A linker, comparing it to a similar ADC except with an enzyme-labile CL2E linker.^[163] The ADC with a CL2A linker showed higher tissue penetration and faster intracellular drug release, increasing DNA damage both in vitro and in vivo. Their study highlighted the advantages of the less stable CL2A linker over the more stable CL2E linker with respect to the bystander effect. Thus, linker design, especially the choice of drug release trigger, substantially impacts ADC bystander effects and must be thoroughly considered during ADC development.

6. Outlook of Linker Design for Next-Generation ADC

Inspired by the clinical success of the approved ADCs, development of next-generation ADC with high performance has been a hot topic in pharmaceutical industry for long. Next-generation ADC is expected to overcome a bunch of challenges on efficacy and safety, confronted by current ADC drugs, such as limited precision of targeting, drug resistance, heterogeneity of constructs, and low DAR.^[164,165] Tethering the payload to the antibody, linker for the next-generation ADC has to be delicately designed for its impact on the PK, plasma stability, and drug release.^[165] To facilitate the goal achieved, we propose that the three aspects to consider for linker design: site-specific bioconjugation, selection of drug release trigger, and spacer optimization. One challenge for ADC development is the heterogeneity of constructs, causing problem to consistency of DAR in manufacturing. By employing site-specific bioconjugation (chemistry, enzyme, or genetic manipulation as discussed in Section 3) in linker design, precise control of DAR can be achieved, guaranteeing high batch-to-batch consistency, providing a reliable foundation for clinical dose control. The drug release trigger for ADC is pivotal for both efficacy and safety of ADC. An ideal trigger must maintain absolute stability in circulation to avoid premature payload release and associated systemic toxicity, while readily responsive to certain conditions (e.g., enzyme, acid, GSH as discussed in Section 2) in the target cells or TME. This “dual-security” mechanism can significantly influence the therapeutic index of ADC. In addition to the drug release trigger, the spacer (mostly self-immolative as discussed in Section 4.2) with attachment to payload is in need of chemical optimization for precise modulation of drug release kinetics, balancing rapid onset with sustained action. Further spacer optimization includes 1) the introduction of hydrophilic groups (e.g., PEG, fleximer, as discussed in Section 4.1) to reduce ADC aggregation, addressing long-standing challenges in formulation stability; 2) design of branched linker architectures allowing co-delivery of synergistic payloads (e.g., cytotoxic drugs combined with immune modulators) to maximize the efficacy or fight against the drug resistance. We believe that linker design incorporating these advancements will empower

next-generation ADCs to overcome critical challenges in cancer therapy.

7. Summary

In summary, ADCs serve as targeted cancer therapies by combining a cancer-targeting antibody with a cytotoxic drug, using a linker to connect these components. The linker comprises a drug release trigger, a bioconjugation group, and spacers that bridge the two components. Linkers are categorized as cleavable or non-cleavable based on whether they include a chemical trigger for conditional drug release. Designing an effective ADC linker requires careful tuning of the drug release trigger, bioconjugation group, and spacer to maximize drug efficacy. This review examines the linkers used in approved ADCs and explores advancements in chemical triggers, bioconjugation, and spacers. We discuss various drug release triggers, such as enzyme, acid, redox-labile triggers. Additionally, we cover bioconjugation strategies, including those involving cysteine and lysine, as well as a brief introduction of the emerging site-specific bioconjugation methods. Additionally, we consider the role of spacers in optimizing linker hydrophobicity, achieving drug release, and branching the linker. Finally, we propose outlook into the linker design for next-generation ADC. With these advancements in linker technology, the linker will continue to play a vital role in ADC development and may eventually contribute to the end of chemotherapy as we know it.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Yaxin Lei: writing—original draft (lead). **Minglei Zheng:** writing—review and editing (supporting). **Peng Chen:** writing—review and editing (supporting). **Chen Seng Ng:** writing—review and editing (supporting). **Teck Peng Loh:** conceptualization (lead); supervision (lead); and writing—review and editing (supporting).

Huitao Liu: conceptualization (lead) and writing—review and editing (lead).

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